

LAND SURFACE TEMPERATURE ANALYSIS OF KERALA FOR TWO DECADES USING MODIS DATA

A dissertation submitted to Pondicherry University in partial fulfilment of the
requirements for the award of degree of

Master of Science in Environmental Sciences

By

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DECLARATION

I, Shijil TC, here by declare that this dissertation entitled “***LAND SURFACE TEMPERATURE ANALYSIS OF KERALA FOR TWO DECADES USING MODIS DATA***”, submitted to Department of Ecology and Environmental Sciences, School of Life Sciences, Pondicherry University, Puducherry in partial fulfilment of the requirements for the award of degree of Master of Science in Environmental Sciences is my original work. I confirm that it has not been submitted, in a part or in a whole, for any other degree or qualification at this or any other academic institution. All sources of information, data, and assistance received during the preparation of this thesis have been duly acknowledged in the text and referenced accordingly. This work represents my own research and analysis unless otherwise stated.

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Certificate

This is to certify that the dissertation entitled “***Land Surface Temperature Analysis of Kerala for Two Decades using MODIS data,***” submitted to the Department of Ecology and Environmental Sciences, School of Life Sciences, Pondicherry University, in partial fulfilment of the requirements for the award of the Degree of Master of Science in Environmental Sciences, is a record of original research work carried out by **Mr. Shijil TC** (Reg. No. 23411027) under my supervision and guidance. This dissertation has not previously formed the basis for the award of any degree, diploma, fellowship, or any other similar title.

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Abberviations

LST : Land Surface Temperature

GEE : Google Earth Engine

NDVI : The Normalized Difference Vegetation Index

NDBI : Normalized Difference Built Vegetative Index

NDWI : Normalized Difference Water Index

TVX : Temperature-Vegetation Index

DEM : Digital Elevation Model

SRTM : Shuttle Radar Topography Mission.

RMSE : Root Mean Square Error

ASTER : Advanced Spaceborne Thermal Emission and Reflection Radiometer

LU/LC : Land Use , Land Cover

SZA : Solar Zenith Angle

TES : Temperature-Emissivity Seperation.

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CHAPTER 1

1.INTRODUCTION

Land Surface Temperature (LST), defined as the radiative skin temperature of the Earth's surface, is a critical variable for understanding energy and water exchanges at the land-atmosphere interface. It serves as a key indicator of environmental health, influencing processes such as evapotranspiration, soil moisture dynamics, and urban heat island effects (Anderson et al., 2008; Karnieli et al., 2010). LST is a fundamental component of the Earth's energy balance, influencing hydrological cycles, microclimates, and ecosystem health. Solar radiation, the primary energy influx, varies with latitude, cloud cover, and atmospheric aerosols, shaping LST distribution (Jacobson, 2000). Vegetation plays a dual role: it cools surfaces through evapotranspiration while enhancing energy absorption via low albedo (Lee et al., 2011; Nemani et al., 1996). Its spatiotemporal variability is driven by the heterogeneity of environmental factors such as solar radiation, cloud cover, vegetation, topography, and anthropogenic modifications (Jacobson, 2000; Mildrexler et al., 2011). Remote sensing has emerged as a pivotal tool for monitoring LST, particularly in data-sparse regions like Kerala, where traditional meteorological networks are limited (Prihodko & Goward, 1997; Czajkowski et al., 2000). Over the past two decades, remote sensing has emerged as an indispensable tool for monitoring LST, particularly in regions like Kerala, India, where meteorological station density is sparse and tropical cloud cover complicates traditional data collection (Prihodko & Goward, 1997; Czajkowski et al., 2000).

Kerala, a biodiverse state in southwest India, Due to its complex terrain, which includes the Western Ghats mountain range, tropical monsoon climate, and rapid land-use changes brought on by urbanization, deforestation, and agricultural growth, it possess a unique case study. These factors significantly alter surface energy balance, making LST analysis vital for assessing ecological and climatic impacts (Li et al., 2015; Mildrexler et al., 2018). Significant environmental pressures, such as shifting vegetation patterns, warming temperatures, and extreme weather events like the 2018 floods, have affected Kerala between 2004 and 2024. These occurrences highlight the necessity of a thorough spatiotemporal LST study.

Land surface temperature (LST) and near-surface air temperature (Tair) are influenced by a combination of atmospheric processes, land surface characteristics, and solar energy. The main source of energy is solar radiation, whose distribution is influenced by geographic latitude, which regulates daily and seasonal variations in sunlight duration and intensity, as well as cloud cover and airborne particulates that scatter or absorb incoming radiation, reducing surface heating. Surface features also have a significant impact: vegetation's darker surfaces (low albedo) absorb more solar radiation, facilitating heat exchange with the atmosphere, while evapotranspiration releases latent heat, lowering temperatures. By changing the rates of soil moisture evaporation and plant transpiration, vegetation density, as measured by the Leaf Area Index (LAI), further influences energy distribution. Temperature fluctuations are

stabilized by soil moisture's ability to retain heat, and the land's ability to emit thermal radiation is determined by surface emissivity, which is controlled by material characteristics. Regional diversity is introduced by topography because features of the terrain, slope direction, and elevation gradients affect airflow patterns and solar exposure, resulting in microclimates. Complexity is increased by atmospheric interactions: clouds, a recurring problem in humid areas, mask satellite-based LST readings by reflecting surface emissions and substituting them with cloud-layer temperatures. Water vapor further distorts temperature data by absorbing and reradiating infrared energy, which makes remote sensing more challenging. Together, these factors result in intricate regional and temporal variations in air and surface temperatures, underscoring the need for diversified approaches to the analysis and prediction of LST-Tair dynamics.

Satellite-derived LST products, such as those from the Moderate-Resolution Imaging Spectroradiometer (MODIS) onboard NASA's Terra and Aqua platforms, provide high-quality, spatially continuous data at 1 km resolution (Wan, 2008; Wan et al., 2002). MODIS algorithms, including the generalized split-window and day/night methods, correct for atmospheric interference and surface emissivity variations, enabling reliable LST estimation even in heterogeneous landscapes (Wan & Dozier, 1996; Wan & Li, 1997).

1. Split-Window Algorithm : Corrects atmospheric interference using thermal bands (31 and 32) to generate 1 km resolution LST.
2. Day/Night Algorithm : Leverages diurnal thermal contrasts to estimate emissivity at 5–6 km resolution (Wan & Dozier, 1996; Wan & Li, 1997).

The MOD11A1 product uses the Moderate Resolution Imaging Spectroradiometer (MODIS) to provide daily global Land Surface Temperature (LST) data as part of NASA's Terra satellite project. This Level 3 dataset, which is structured in a sinusoidal grid system with tiles that span around 1200 x 1200 km, is essential for climate research and environmental monitoring since it captures thermal dynamics at a spatial resolution of 1 km (nominal at nadir). Although the program provides daily observations, there may be gaps because to sensor limitations or cloud cover. Thermal infrared bands 31 and 32 are used to produce the daytime and nighttime LST values (in Kelvin), together with emissivity data for these bands, quality control (QC) layers to evaluate dependability, and metadata such observation timing. The retrieval approach uses a generalized split-window method that uses emissivity estimations from land cover datasets such as MOD12Q1 to account for air interference. Variability in surface conditions and cloud cover can affect precision, even though the device is intended for ± 1 K accuracy in clear skies. MOD11A1, encoded in HDF-EOS format, has provided global coverage since February 2000. Among its applications are hydrological modeling, climate trend analysis, wildfire risk assessment, agricultural drought monitoring, and urban heat island studies. Users are advised to fill in the gaps in cloud-related data and remove low-confidence pixels by using QC flags. The MOD11A2 8-day composite product is an additional resource for more consistent coverage. The data can be

accessed via NASA's LP DAAC or Earthdata Search, and mosaicking tools such as GDAL or AppEEARS are sometimes required for regional or global analysis. Overall, MOD11A1 is crucial for understanding surface thermal processes, but its applicability depends on how well quality metadata and cloud-related limitations are managed. Cloud contamination remains a major challenge, especially in Kerala's humid tropics, where persistent cloud cover obscures land surfaces and introduces errors in nighttime retrievals (Czajkowski et al., 2000; Vogt et al., 1997).

Kerala's environmental dynamics—such as its abundant greenery, monsoonal rainfall, and rapid urbanization—make it a desirable location for LST research. According to LST trends, the state has seen increased urban heat islands (UHI), deforestation, and extreme weather events (such as the 2018 floods) from 2004 to 2024. In contrast to urban areas where impermeable surfaces enhance LST, vegetation has a cooling effect that is driven by surface roughness and transpirational cooling. By integrating LST data with land cover and climate variables, this study closes information gaps regarding Kerala's thermal environment. It highlights the ways in which remote sensing might support the attainment of sustainable development goals, such as lowering heat stress and preserving ecosystems (Albright et al., 2011; Mildrexler et al., 2018). As LST continues to gain traction in climate science, this study emphasizes its significance in addressing regionally specific environmental challenges.

Kerala's land surface temperature will be examined for 20 years in this project. Here, the time period will be divided by the sample year interval of five years, for a total of twenty years. Thus, we will examine the LST for the following years: 2004, 2009, 2014, 2019, and 2024, specifically. MODIS data are obtained from the MOD11A1 satellite.

1.1.OBJECTIVE

1. To analyse the LST of Kerala by using MOD11A1 latitudinal wise (8° - 13° latitude), for the years of 2004, 2009, 2014, 2019, 2024
2. To analyse the LST of Kerala altitudinal wise (Low land, Mid land, High land) using MOD11A1, for the years of 2004, 2009, 2014, 2024.

CHAPTER 2

2.REVIEW OF LITERATURE

Understanding land surface temperature is crucial for comprehending ecological changes, urban heat islands, and climate dynamics. Due to deforestation, urbanization, and agricultural expansion, Kerala, a tropical coastal region at the bottom of India, has undergone significant changes in land use. The research on remote sensing of the LST in Kerala from 2004 to 2024 is the primary emphasis of this work.

Significant progress has been made in satellite-derived LST recovery, including techniques that are adapted to surface conditions and sensor capabilities. However, there are still challenges to improve accuracy and broaden the applicability of LST interpretation, validation, and atmospheric correction on a variety of surfaces. The focus of future research should be on multi-sensor fusion, hyperspectral data, and improved physical models. Emerging technologies like neural networks and advanced RTMs have the ability to overcome current limitations.

The field of remote sensing is becoming increasingly significant, aiding in many areas of research and serving as a crucial instrument for determining air temperature and LST over wide geographic ranges. Therefore, this review of the literature will examine the major works that investigate the connection between elevation model research and MODIS Land Surface Temperature (LST) data, using various approaches to obtain precise LST estimation.

KEY STUDIES AND METHODOLOGIES

2.1. Benali et.al (2012); Statistical modeling for weekly Ta estimation in Portugal.

Benali used MODIS LST data and auxiliary factors during decades (2000-2009) to estimate the weekly (Tavg), maximum (Tmax), and minimum (Tmin) air temperatures throughout Portugal. They have optimized using sophisticated statistical models, such as the **Jackknif** approach and the **Bootstrap** method, in order to take sparse station data into consideration. using MODIS Terra LST (night/day) data. In statistics, resampling techniques called bootstrap and **Jackknif** tools are used to quantify the variability and possible bias of a statistic or estimator. Both entail creating new data sets from the original data in order to assess bias, standard error, and confidence intervals.

The key findings they found was;

High accuracy for Tavg (RMSE = 1.33°C, MEF = 0.941) is attained. They discovered that LST nighttime data performed better for Tavg estimate than daytime data. Because of the fog and damp sea wind, coastal areas also showed greater mistake rates. Among the study's limitations are the underestimation of annual Ta extremes and the decline in performance during cloudy months.

2.2. Zhu et.al (2013);

TVX Method for daily Tmax & Tmin in the Tibetan plateau

The primary goal of this study was to use the MODIS LST & TVX method to determine the daily Tmax and Tmin in the Xiangride river basin. Here, they used the Temperature Index, or TVX, approach. By assuming a negative correlation between LST and NDVI and extrapolating Ta for fully vegetated pixels (NDVI Max = 0.86), they obtained the MODIS Terra/Aqua LST, NDVI, and station data using the TVX (The Temperature Index) approach. TVX enhanced Tmax estimation (RMSE=3.79 vs. 7.45 degrees Celsius without TVX), and they were able to accurately estimate nighttime LST using MODIS data. The study has certain limitations, such as the TVX method's failure in higher altitude regions (like Maduo) because of a positive LST-NDVI correlation and the data's decreased availability owing to cloud cover.

2.3. Anjali & Roshini (2022), John et.al (2021)

Land Surface Temperature Dynamics And It's Relationship With Land use/Landcover (LULC) Changes

According to the study, urbanization and deforestation greatly raise LST because they decrease evapotranspiration and increase albedo, which causes a heating impact. Kerala lost around 12.65% of its forest, which caused the LST to climb (from a minimum of +0.9 degrees Celsius to a maximum of +5 degrees Celsius) and the urban area to expand by 14.9%. 77% of the Bharathapuzha river basin is now in higher LST zones as a result of an increase in urban area and barren land of 2.23% and 39.5%, respectively.

Methodologies they have used are;

1.Statistical Tool; Mann-kendrall trend tests and breakpoint analysis have used to analyse the reveal non-linear LST trends with significant breaks.

2.Remote sensing & GIS; Both Landsat TM/ETMI/OLI and MODIS data were used to derive the LST by 2 methods, 1) Mono-Window algorithms and Spectral radiance method.

The lack of better resolution temporal data resulting from capturing diurnal/seasonal LST fluctuations and the restricted integration of socioeconomic variables with the biophysical method are the method's limitations.

2.4. Thomas et.al (2015)

Digital Elevation Model and Their Application in Topographic Analysis

The representation of the earth's topography using digital elevation models is a crucial tool that is utilized in many different domains, including environmental modeling, hazard mapping, hydrology, and geomorphometry. In this work, Thomas et al. (2015) assess whether freely accessible global DEMs (ASTER, SRTM, and GMTED) are appropriate for topographic investigation in tropical areas. in contrast to a high-resolution digital elevation model that is based on topographic maps with a scale of 1:50,000.

They examined vertical accuracy, accuracy-affecting factors, geomorphometric derivatives, and DEM performance in order to compare the accuracy of the DEM. They discovered that TOPO has the lowest error and SRTM has the highest vertical accuracy (RMSE = 17.05). (RMSE=13.17M), and DEM inaccuracy is strongly influenced by terrain, roughness, and vegetation cover. Because of sensor limitations, RMSE is higher in forested and high relief areas.

Some of the better DEMs are freely accessible, such as Ithikkara (Kerala) and the tropical river basin. However, ASTER is a good substitute for contour-based DEMs and for finer-scale investigations. High precision is needed, and future research should concentrate on methods to improve the accuracy of DEMs in challenging terrain.

2.5.Arul Balaji, Padmalal and Maya (2020)

Urbanisation And Land Surface Temperature Changes on Coastal Area's

Here in the study, they examine the impact of urbanization on the LST Of Thiruvananthapuram city using remote sensing and GIS technique. The main reason for increase in the LST in Thiruvananthapuram are, Urbanization and environmental degradation, Land use / Landcover changes, Urban heat island effect,etc. Within the time span of 1988-2019, nearly 10km² of built-up area has increased in Thiruvananthapuram, and also 125Km² to 71Km² of vegetation cover also has declined. Rapid urbanization happened particularly in

tropical regions and subtropical regions, led to large scale environmental degradation, including vegetation loss, increased impervious surfaces and contamination of waterbodies.

According to a study, Thiruvananthapuram's average LST increased by 1.6 degrees Celsius between 1988 and 2019. Urban heat islands also have a significant effect on LST. For the LU/LC alterations and LST variation, the researchers in this study employed remote sensing indices like the Normalized Difference Vegetative Index (NDVI), Normalized Difference Built Vegetative Index (NDBI), and Normalized Difference Water Index (NDWI). This study highlights the difficulties of urbanization-induced warming and is in line with international research like those conducted in Nairobi (3.65 degrees Celsius LST rise) and Hanoi (7.5 to 10.8 degrees Celsius LST increase). Overall, this study finds that urbanization will change a place's LU/LC and cause natural ecosystems to deteriorate. One important effect of urbanization is UHI, which will raise quantifiable LST. Sustainable urban planning including Green Infrastructure is essential to counteract environmental degradation.

2.6.Satheendran et.al (2022)

Urbanization and Land Surface Temperature Changes in Coastal Areas

This study uses remote sensing to examine how urbanization affects LST in Karunagappally, a rapidly expanding coastal town in Kerala. Rapid urbanization causes major environmental deterioration, including the loss of vegetation, the conversion of arid lands, and an increase in impermeable surface, particularly in tropical coastal regions.

The accumulation areas in Karunagappally are a major contributor to environmental change; between 1988 and 2018, they grew from 1% to 22% of the entire research area. Another important factor influencing the variation in LST is the urban heat island effect. According to a study, Karunagappally's LST rose from 22.2 to 29.2 degrees Celsius between 1988 and 2018. Here in this study also, they have used remote sensing indices such as , NDVI, NDBI, and another effective tools to analyse and monitor the LST effectively.

The study highlights that in order to reduce UHIs, green cover must be increased and urban planning must be appropriate. This research also compares and aligns its findings with those of other researchers throughout the world, including those in Hanoi (7.5–10.8 degrees Celsius LST rise) and Nairobi (3.65 degrees Celsius LST increase). According to the study's findings, urbanization frequently results in major LU/LC alterations at the expense of natural ecosystems. Urbanization and sustainable urban planning, which includes green infrastructure as a means of combating environmental

deterioration, are major contributors to UHI, which also has a significant impact on LST.

2.7. Mostovoy et.al (2006)

Tmax and Tmin Estimation of Mississippi Using Remote Sensing.

In this case study, they used MODIS Land Surface temperature data from the Mississippi region (2000-2004) to estimate the daily maximum ($T_{a\ max}$) and minimum ($T_{a\ min}$) using statistical methods. The study assesses the following factors that affect the Correlation Coefficient (R) between LST & T_{air} : pixel resolution ($1\ km^2 \times 5\ km^2$), satellite overpass time, season, landcover, vegetation fraction, and view zenith angle. The studies' main conclusions are that, in comparison to coarser data ($5\ km^2$), higher resolution ($1\ km^2$) should enhance correlation. According to Norman et al. (1995), the mixed-pixel effect, in which soil and vegetation produce distinct heat signals, is responsible for the linear correspondence with Zenith angle (1° – 2° from 0° to 65°).

The forest average maximum was lower than the grassland average ($-0.2^\circ C$ to $-0.8^\circ C$), while the nighttime average minimum exhibited no angle dependence ($-1.5^\circ C$ to $-1.8^\circ C$). R For T_{max} -LST decreased to 0.6-0.9 in summer but stayed constant (-0.9) in winter due to the moderating influence of vegetation modeling on Surface Temperature (Gallo C et al. 1993) and seasonal variability, which controlled "R" swings more than land Cover type. According to Nyazikhin et al. (1978), the vegetation fraction derived from MODIS LAI influenced the average but showed a weak direct correlation with R. The studies' practical implications include the discovery of improved spatial interpretation accuracy of MODIS LST (1 - $2^\circ C$) for T_{a} fields, especially in clear sky conditions and energy balance models Verified average with minimal contributions from spatial covariations, controlled via surface energy partitioning.

2.8. Duan et.al (2014)

Time - Consistent LST Product from Modis Data;

The primary goal of this research is to create a technique for normalizing Terra-MODIS LST to a consistent local solar time (11:00) in order to resolve temporal discrepancies brought on the MODIS 3 scanning characteristics. They have employed a linear regression model to estimate this. LST versus local solar time (thus the slope) (10:00–12:00). according to MSG-BEVIRI's digital elevation model (DEM), Solar Zenith angle (SZA), and NDVI data. After applying the regression on MODIS LST for temporal normalization, they verified the results against MSG-SEVIRI LST and discovered that the RMSE decreased from 1.5k

(before normalization) to 0.5k (after normalization). Additionally, robusters were demonstrated throughout the study zones, land cover types, and several months. The study's shortcomings include assuming stable connections between predictors and LST, ignoring cloud contamination and angular impact, and assuming linear LST changes between 10:00 and 12:00.

2.9. Zhao-Liang Li et.al (2012)

Satellite Derived Land Surface Temperature (LST)

This research examined the state of satellite-derived LST retrieval, including its methods, difficulties, and potential future directions. Additionally, various retrieval techniques for LST analysis are mentioned.

Single-Channel Techniques: Need a known LSE and precise atmospheric adjustment. As an example, consider Jiménez-Muñoz and Sobrino's (2003) generalized single-channel method, which works well with little input data but has trouble with high ambient water vapor.

Split-Window (SW) algorithms: To account for atmospheric effects, use differential absorption in two neighboring TIR channels (e.g., 11 μm and 12 μm). Although it necessitates precise LSE estimations, the Generalized Split-Window (GSW) approach by Wan and Dozier (1996) is popular due to its resilience.

Multi-Angle Methods: Limit atmospheric dependence by utilizing observations from many angles, such as ATSR data. Although these approaches are effective, they are constrained by geometric registration issues and differences in angular emissivity.

Temperature-Emissivity Separation (TES): Used in hyperspectral sensors like ASTER, TES relies on spectral contrast to separate LST and LSE. It performs well for high-emissivity contrast surfaces but is less accurate for grey bodies (Gillespie et al., 1998).

2.10. Weng et.al (2014)

High Resolution Daily LST By Fusing Landsat And MODIS Data

Here they proposed a **Spatial Temporal Adaptive Data Fusion Algorithms** for temperature (SADEAT) to generate daily LST at Landsat resolution (120 M) by blending Landsat and MODIS data. They have adopted Enhanced STARFM model by Incorporating the Annual temperature cycle (ATC) and liners spectral mixture analysis. (LSMA) to account for urban thermal

heterogeneity. also used Landsat's reflective and thermal bands to identify spectrally similar pixels and weighted their contributions. and they found Achieved mean absolute differences of 1.3-2.0 K between Predicted and observed LST's also preserved spatial detail's (eg; roads, urban features) and seasonal LST variations. The limitations of the study are, they Assumes the stable phase shifts in ATC, limiting land prediction accuracy for unrecorded LST change (eg; land cover change) and also requires careful parameter tuning.

Temperature Variations and Land Cover: Due to mixed-pixel thermal emissions from soil and plant, daytime temperature variations ($\Delta_{\max} = T_{a\max} - \text{LST}$) rose linearly with MODIS view zenith angle ($1-2^{\circ}\text{C}$ over 0° to $\pm 65^{\circ}$) (Norman et al., 1995). There was very little angle dependence in nighttime variations ($\Delta_{\min}$). As a result of vegetation's moderating effect on surface temperatures, forests showed smaller $\Delta_{\max}$ (-0.2°C to -0.8°C) than croplands or grasslands (-1.5°C to -1.8°C) (Gallo et al., 1993).

Seasonal and Vegetation Effects: Seasonal weather patterns, particularly cold fronts in October–April, drove R variability more than land cover type. Summer correlations ($R = 0.6-0.9$ for $T_{a\max}$ -LST) were less than winter values (~ 0.9), perhaps due to atmospheric instability. Though it had little direct correlation with R, the MODIS-derived leaf area index (LAI) (Knyazikhin et al., 1998) affected Δ magnitudes, indicating that sub-pixel heterogeneity or other factors were at play.

Applications and Validation: For mapping air temperatures under clear skies, MODIS LST increased spatial interpolation accuracy by $1-2^{\circ}\text{C}$. While 1.0 km^2 data improved local precision, the 5.0 km^2 product (MOD11C1) provided wider coverage. It was proven by energy balance models (Gao, 1995) that surface energy partitioning (such as solar radiation and latent heat) rather than spatial covariations controlled Δ .

With suggestions to give resolution priority for local applications and seasonal tweaks for model correctness, the study highlights MODIS LST as a dependable tool for regional temperature estimation. These discoveries bolster agricultural and ecological models that need temperature inputs with great resolution.

CHAPTER 3

3.METHODOLOGY

MODIS sensors were launched on board the National Aerodynamics and Space Administration (NASA) Earth Observing System (EOS) **Terra and Aqua satellites in December 1999 and May 2002**, respectively ([Jang et al., 2010](#)). It provides daylight and nighttime LST readings, emissivity estimates, quality control flags, and observation durations in a sinusoidal grid projection, all derived from data made by the Terra (since 2000) and Aqua (since 2002) satellites. **The Terra overpass time is around 10:30 AM (local solar time) in its descending mode and 10:30 PM in its ascending mode.** The Aqua overpass time is around 1:30 PM in its ascending mode and 1:30 AM in its descending mode ([Wang et al., 2006](#)). This dataset is essential for applications that record surface thermal characteristics and diurnal temperature fluctuations, including climate modeling, urban heat island study, agricultural monitoring, and environmental research. It offers daily emissivity and land surface temperature (LST) data with a spatial precision of one kilometer. **MOD11A1** is accessible through **NASA's Land Processes Distributed Active Archive Centre (LP DAAC)** or **USGS Earth Explorer**, requiring specialized tools for processing.

3.1. ArcGis

The ArcGIS software system from ESRI is a complete GIS software package. This integrated platform consists of several components and is designed for the creation of functional GIS solutions. It includes tools for handling and storing geographic data in files and databases, as well as a geographic information model for modeling real-world occurrences. This versatile software simplifies a wide range of geographic tasks, from data production and editing to integration, management, modification, analysis, mapping, and reporting, making it user-friendly. Furthermore, ArcGIS Online extends its capabilities with web services that are accessible via web-enabled devices, browsers, and applications (Maguire, 2008).

3.2. KERALA

Kerala is a state in southwest India that is tucked away between the Western Ghats and the Arabian Sea. **Situated between latitudes 8 and 13° and longitudes 74° 40' to 77° 50'**, it makes up 1.18% (38,863 km²) of the nation's total land area. Because of its distinct geology, biodiversity, and climate variability, it is a dynamic study location for environmental and climatic research. Its topography includes coastal plains, midland hills, and the biodiverse Western Ghats mountain range, a UNESCO World Heritage Site. The Western Ghats in the east, which make up around 640 km of the state,

and the Arabian Sea in the west, which gives the state a 590 km coastline, serve as the state's boundaries.



Plt.1 Kerala Physical Map

3.3. GOOGLE EARTH ENGINE

Google Earth Engine (GEE) is a planetary-scale platform for Earth science data and analysis ([Gorelick et al., 2017](#)) Google Earth Engine (GEE; [31]) is an online platform created to allow remote sensing users to easily perform big data

analyses without the need for computation resources. It has a multi-petabyte catalog of satellite imagery and geospatial dataset with planetary-scale analysis capabilities which are available for users to detect changes, map trends, and quantify difference on the Earth's Surface ([Gorelick et al., 2017](#)). All MODIS data sets are directly available to GEE.

The LP DAAC (NASA Earth Engine) offers the MODIS data (MOD11A1) for download. Therefore, we must perform spatiotemporal analysis of the data in order to analyse the temperature from the MODIS data for the LST analysis of Kerala. Therefore, the first step is to import the data in HDF format into ArcGis software. Twelve layers will make up this **HDF** file; **layer 0 will be utilized for daytime LST analysis, and layer 4 will be used for nocturnal LST analysis**. The LST of the provided data can be automatically determined by employing this spatiotemporal analysis.

3.4. SPATIOTEMPORAL ANALYSIS

We must choose the work folder from the **Catalog** and then pull out the "**toolbox**" for that work folder in order to perform the spatiotemporal analysis. The model has been created from that toolbox. A few steps are taken from the model to determine the temperature in degrees Celsius. The steps are given in the fig.1 below.

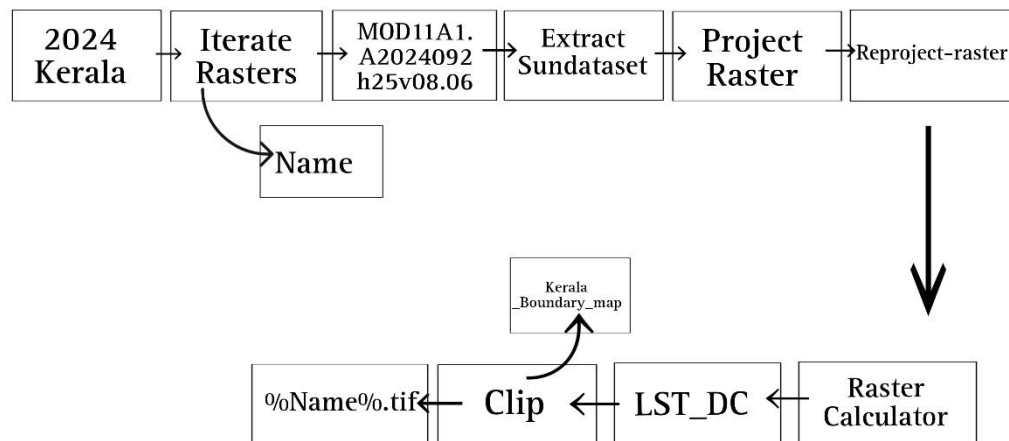


fig.1

The LST in the raw data will be in the Kelvin scale, which we may convert using a "**Spatial Analysis Tool**". By using **Raster Calculator**, we can convert

the **Kelvin value in to Degree Celsius**, which is given below in fig.2. Following that, it can be saved and executed. After a few minutes, the automatically extracted data will be prepared for analysis.

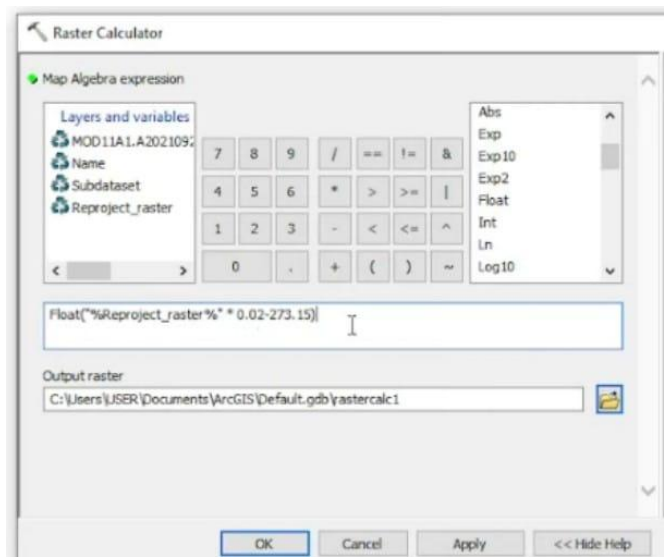


fig.2

Therefore, it is quite challenging to analyse all of the data in this study in this way. We are analysing about 20 years of data here, with 5-year intervals between 2004–2009, 2014–2019, and 2024. Therefore, it takes a lot of time. Additionally, we must analyse the average monthly temperature for each interval year; Google Earth Engine is a preferable choice for this. Here, we can enter code that is appropriate for our needs, and it will analyse and extract the information. The code script for analysing Kerala's LST was obtained from ChatGPT and loaded into GEE to analyse the monthly average temperature of a few chosen years using MOD11A1 data. Additionally, this data analysis can be exported as a TIFF file, which facilitates additional study.

Here in this study, the analysis of LST has approached in 2 ways;

1. Latitudinal Analysis

2. Altitudinal Analysis

3.5.1. Latitudinal Analysis

Code script used for the **Latitudinal Wise Analysis** (Dates are modified according to the time period);

```
// Define the region of interest: Kerala, India
var kerala = ee.FeatureCollection('FAO/GAUL/2015/level1')
               .filter(ee.Filter.eq('ADM1_NAME', 'Kerala'));

// Load the MOD11A1 dataset
var dataset = ee.ImageCollection('MODIS/061/MOD11A1')
               .filterDate('2004-01-01', '2004-01-30')
```

```

        .filterBounds(kerala);

// Select the daytime LST band and convert from Kelvin to Celsius
var lstCelsius = dataset.select('LST_Day_1km')
    .map(function(image) {

        return
        image.multiply(0.02).subtract(273.15)
        .copyProperties(image,
        ['system:time_start']));

});

// Calculate the mean LST over January 2004
var meanLST = lstCelsius.mean().clip(kerala);

// Define visualization parameters
var vizParams = {
    min: 20,
    max: 40,
    palette: ['green', 'yellow', 'red']
};

// Add the mean LST layer to the map
Map.centerObject(kerala, 7);
Map.addLayer(meanLST, vizParams, 'Mean LST January 2004');
// Export the mean LST image to Google Drive
Export.image.toDrive({
    image: meanLST,
    description: 'Kerala_Mean_LST_Jan_2004',
    folder: 'EarthEngineExports', // Optional: specify a folder in
your Drive
    fileNamePrefix: 'Kerala_Mean_LST_Jan_2004',
    region: kerala.geometry(),
    scale: 1000, // Export at 1 km resolution
    crs: 'EPSG:4326', // Set the coordinate reference system to WGS84
    maxPixels: 1e13 // Increase limit for large exports
});

```

Kerala is located approximately 8°18'N to 12°48'N. So, we can classify Kerala into 5 classes according to latitude.

1. 8°–9°N
2. 9°–10°N
3. 10°–11°N
4. 11°–12°N
5. 12°–13°N

For further analysis, the research region must be divided into five distinct classes; a **clipping** tool has been employed for this purpose. The **layout view** must be enabled in order to obtain the longitudinal values on the map. such that it is simple to cut the image in accordance with the longitudinal classes after obtaining the longitudinal values. First have to select the work folder from the catalog, then the process has shown by the below flowchart, fig.3.

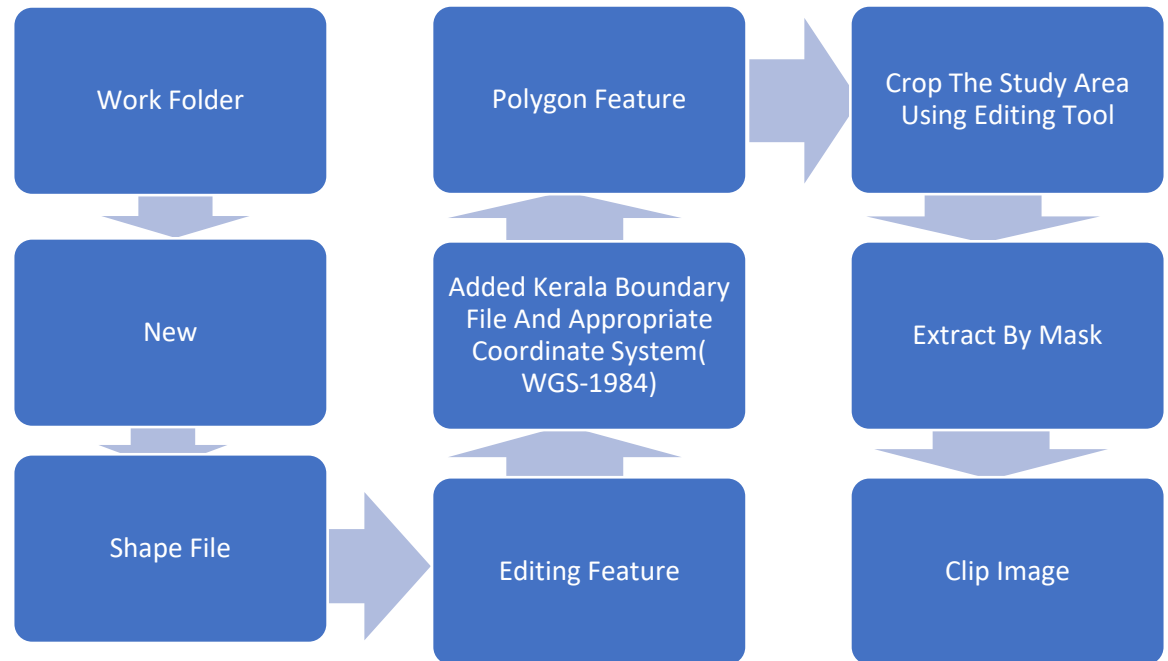


fig.3

After that, each clip that corresponded to each longitudinal class for each year interval was examined. There are five-year intervals (2004, 2009, 2014, 2019, 2024), and for each year, the monthly average has been examined (**from over 300 data sets**), and the monthly maximum and minimum have been extracted from the databases.

3.5.2.Altitudinal Analysis

Code script for **Altitudinal Wise Analysis**, (Dates are modified according to the time period);

```

/ Define elevation classes (in meters)
var elevationClasses = [0, 200, 500, 1000, 1500, 3000];

// Load MODIS LST dataset for January 2004
var modisLST = ee.ImageCollection('MODIS/061/MOD11A2')
  .filterDate('2004-01-01', '2004-01-31')
  .select('LST_Day_1km')
  .mean()
  .multiply(0.02) // Scale factor
  
```

```

        .subtract(273.15); // Convert to Celsius

// Define Kerala boundary
var kerala = ee.FeatureCollection('FAO/GAUL/2015/level1')
    .filter(ee.Filter.eq('ADM1_NAME', 'Kerala'));

// Clip elevation and LST data to Kerala
var elevation = srtm.clip(kerala);
var lst = modisLST.clip(kerala);

// Create elevation classified image
var elevationClassified = ee.Image(0)
    .where(elevation.gte(elevationClasses[0]).and(elevation.lt(elevationClasses[1])), 1)
    .where(elevation.gte(elevationClasses[1]).and(elevation.lt(elevationClasses[2])), 2)
    .where(elevation.gte(elevationClasses[2]).and(elevation.lt(elevationClasses[3])), 3)
    .where(elevation.gte(elevationClasses[3]).and(elevation.lt(elevationClasses[4])), 4)
    .where(elevation.gte(elevationClasses[4]), 5);

// Compute min and max LST for each elevation class
var stats = elevationClassified.addBands(lst).reduceRegion({
    reducer: ee.Reducer.minMax().group({
        groupField: 0,
        groupName: 'Elevation_Class'
    }),
    geometry: kerala.geometry(),
    scale: 1000,
    bestEffort: true
});

// Print the results
print('LST min/max for each elevation class:', stats);

```

Kerala's elevation is not evenly distributed, making it challenging and time-consuming to process in ArcGis. However, GEE made the procedure simple and assisted in processing the elevation-wise LST of Kerala for five interannual years (2004, 2009, 2014, 2019, and 2024). Using suitable code scripts discovered from ChatGpt, data is directly obtained from the Google Earth engine for altitudinal-wise classification. This approach analyses almost **360 samples**.

Scripts generated via ChatGPT classify Kerala's elevation into six categories:

1. Class 1: >0 m
2. Class 2: 0–100 m

3. Class 3: 100–500 m
4. Class 4: 500–1000 m
5. Class 5: 1000–2000 m
6. Class 6: 2000–3000 m

Using these scripts, elevation-specific LST values are extracted for the same five-year intervals (2004–2024), covering 360 datasets. Monthly average temperatures for each elevation class are computed within GEE, streamlining the analysis of Kerala's diverse topography.

This dual-method approach combines ArcGIS for spatial preprocessing and GEE for scalable, automated computation, ensuring comprehensive insights into Kerala's LST variations across both latitude and elevation gradients.

3.6. NDVI Analysis

The Normalized Difference Vegetation Index (NDVI) is a widely used remote sensing indices for the analysis and quantifying of health and density of vegetation. NDVI will be a value between -1 and 1, less than 0 suggest lack of dry land.

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}}$$

$$\text{NDVI} = \frac{\text{Band5} - \text{Band4}}{\text{Band5} + \text{Band4}}$$

Here in this study the NDVI is also derived from the GEE, using the code script;

```
// Load the boundary of Kerala from GAUL 2015 level1 dataset
var kerala = ee.FeatureCollection("FAO/GAUL/2015/level1")
    .filter(ee.Filter.eq('ADM1_NAME', 'Kerala'))
    .filter(ee.Filter.eq('ADM0_NAME', 'India'));

// Center the map on Kerala
Map.centerObject(kerala, 7);

// Load MODIS NDVI (MOD13Q1: 16-day composite at 250m)
var ndvi = ee.ImageCollection('MODIS/006/MOD13Q1')
    .filterDate('2004-01-01', '2004-12-31')
    .select('NDVI')
    .map(function(image) {
        return
image.clip(kerala).multiply(0.0001).copyProperties(image,
image.propertyNames());
    });

// Compute the mean NDVI for the entire year
```

```

var ndviMean = ndvi.mean();

// Visualization parameters
var ndviVis = {
  min: 0.0,
  max: 0.9,
  palette: ['beige', 'yellowgreen', 'green', 'darkgreen']
};

// Add the NDVI layer and Kerala boundary to the map
Map.addLayer(ndviMean, ndviVis, 'Mean NDVI - Kerala 2004');
Map.addLayer(kerala, {color: 'black'}, 'Kerala Boundary');

// Print to console
print('Mean NDVI 2004 (Kerala)', ndviMean);

// Optional: Export to Google Drive
Export.image.toDrive({
  image: ndviMean,
  description: 'Kerala_NDVI_2004',
  folder: 'GEE_Exports',
  scale: 250,
  region: kerala.geometry(),
  maxPixels: 1e13
});

```

CHAPTER 4

4.RESULT AND DISCUSSION

4.1.latitudinal wise

4.1.1. 2004

When Kerala's 2004 temperature and NDVI data are analysed, clear seasonal and regional patterns as well as significant anomalies are shown. The 10–11° latitude had the greatest average temperatures, with extreme highs of 43.17°C in March and 41.43°C in April. This was in spite of a high NDVI value of 0.9748, which is usually linked to vegetation-driven cooling. This discrepancy points to regional causes like urbanization, deforestation, or even data problems. With the highest NDVI (0.9857) and more mild highs (e.g., 41.80°C in April), further high values are in the table below, Tab.1.

2004 Highest Monthly Average LST	8°-9°	9°-10°	10°- 11°	11°-12°	12°-13°
January	33.0191	33.5252	35.0422	32.6536	35.483
February	35.2019	35.7062	39.0616	35.0271	37.93
March	36.71	41.857	43.1662	39.9627	39.3462
April	35.9214	38.33	41.4254	38.375	41.7986
May	34.45	35.63	38.73	34.75	37.86
June	31.39	31.9167	34.31	34.8433	30.83
July	33.17	33.25	36.51	37.3035	29.85
August	32.8433	32.435	38.99	38.1833	29.98
September	37.41	35.31	35.51	39.958	35.12
October	36.518	33.655	33.3867	33.2557	33.015
November	32.374	31.7329	32.7643	31.69	34.3245
December	32.9377	32.5913	32.477	31.8274	34.3275

Tab.1. Highest monthly average of 2004 in degree Celsius.

The 12–13° latitude, on the other hand, was consistent with vegetation's function in stabilizing temperatures. " With implausible extremes like 0.43°C in August (probably a data entry error) and legitimate lows like 3.8°C in November and 7.07°C in September that correspond with monsoon cooling, the lowest temperatures were recorded in the 9–10° latitude. Its strong NDVI further supported the steady lows (~19–24°C) across latitudes 12–13°, highlighting the moderating effect of vegetation, further low values are showed in the table below, Tab.2.

2004 Lowest Monthly Average LST	8°- 9°	9°- 10°	10°- 11°	11°- 12°	12°- 13°
January	21.2679	20.5211	18.7274	20.4638	23.6889
February	23.1037	21.821	19.9683	22.1582	26.0868
December	20.3589	19.0445	19.1357	20.1848	23.6639
March	24.96	24.1167	24.9357	24.9357	27.6637
April	24.115	23.63	20.8971	21.2082	26.315
May	21.41	19.37	10.75	9.57	24.06
June	8.15	8.39	11.93	9.41	18.97
July	12.69	11.77	10.83	7.61	20.85
August	8.53	0.43	8.99	13.03	14.69
September	20.82	7.07	9.87	7.97	22.93
October	22.47	18.455	17.535	17.785	22.3067
November	17.7633	3.8	16.49	17.2589	19.0445
December	20.3589	19.0445	19.1357	20.1848	23.6639

Tab.2 lowest monthly average of 2004 in Degree Celsius

The hottest months of the year were March through April, with continuous highs during the summer months of March through May, especially in the latitudes of 10 to 11°. Valid data suggested mild lows (15–25°C), although the monsoon months (June–September) showed inconsistent lows, including dubious figures as 0.43°C in August. The importance of vegetation was highlighted by the NDVI-temperature relationship: the 12–13° latitude (NDVI: 0.9857) showed stable lows and mild highs, whereas the 10–11° latitude (NDVI: 0.9748) defied expectations with excessive heat, suggesting non-vegetation factors. Higher lows (e.g., 21.26°C in January) and highs (e.g., 36.71°C in March) were recorded in lower NDVI zones, such as the 8–9° latitude (NDVI: 0.87485), which reflected the reduced cooling ability of sparse vegetation. The fig.4 below will be mentioning the NDVI value of 2004.

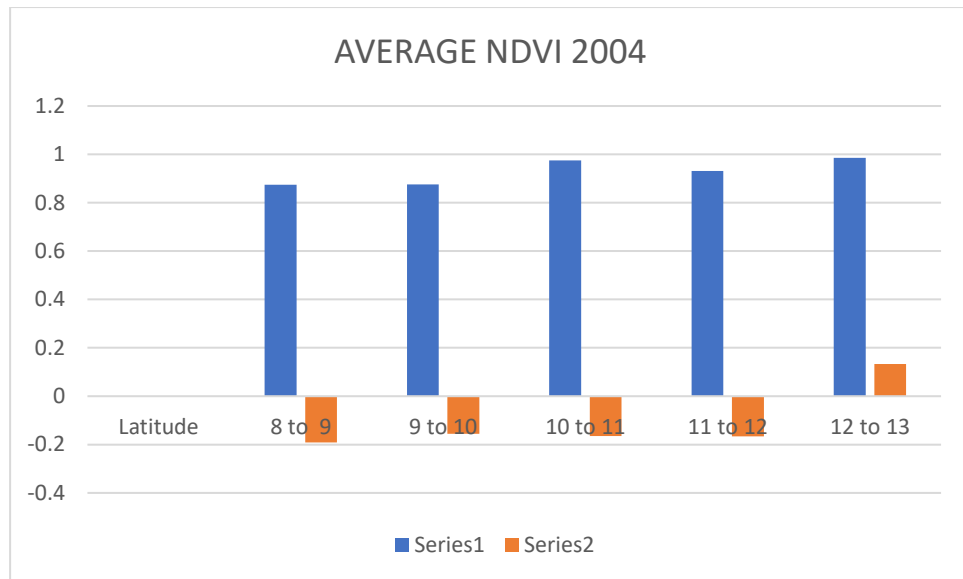
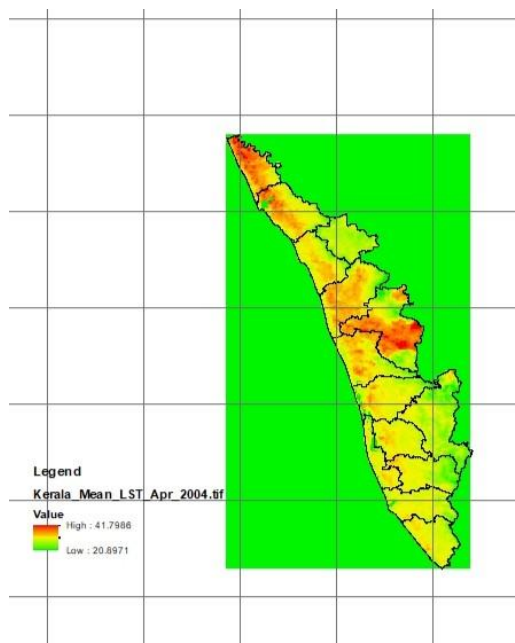


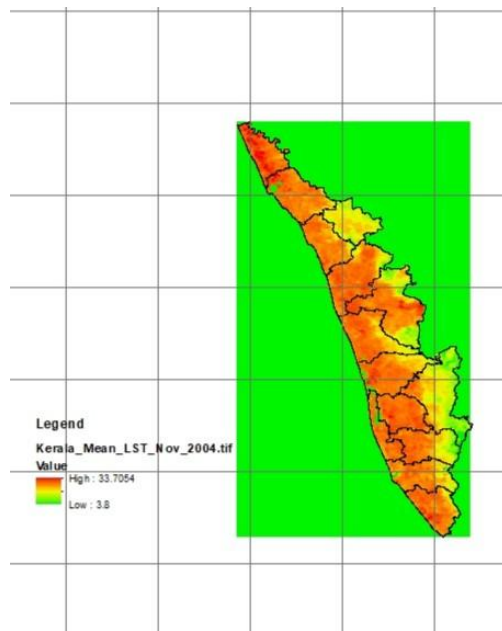
Fig.4

In conclusion, because of its high summer temperatures, the 10–11° latitude was the hottest, whereas the 12–13° latitude struck a balance between warmth and stability brought on by vegetation. Unreliable monsoon data complicated trends, and seasonal variability peaked in the summer. For a comprehensive knowledge of Kerala's climatic dynamics, it is necessary to look into external influences like land-use changes or sensor imperfections. The NDVI's significance in temperature moderation was clear but obscured by anomalies.

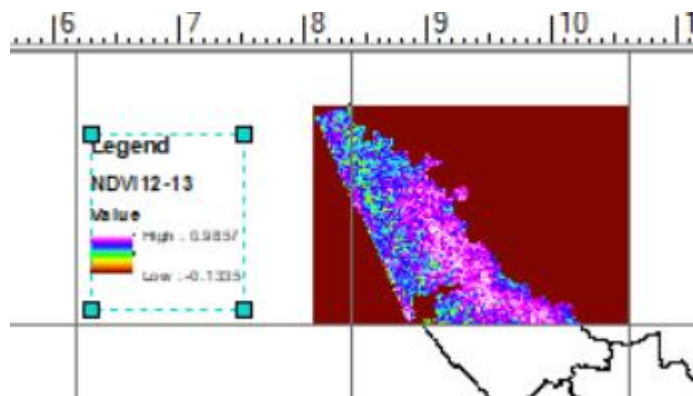
Here are some snapshots of LST analysis of 2004;



plt.2 LST map of Kerala , April 2004.



plt.3 LST map of Kerala, November 2004



pl.4, Clip image of NDVI map of 12° – 13° latitude of 2004.

4.1.2. 2009

Kerala's 2009 temperature and NDVI data study highlights unique thermal patterns influenced by vegetation cover, seasonal cycles, and latitude. During the pre-monsoon summer, the 10°–11° latitude had the highest monthly averages, reaching a peak of 43.65°C in May and 41.29°C in April due to strong solar radiation. Even though the NDVI was moderate (0.8669), suggesting that vegetation was partially responsible for cooling, solar dominance was nonetheless dominant. In contrast, the 12°–13° latitude, which has less vegetation (NDVI: -0.1612), recorded somewhat lower highs (42.26°C in May and 40.16°C in April), indicating less heat retention in areas with little vegetation. Tab.3 will be showing the highest monthly average in 2004.

2009 Highest Monthly Average LST	8°- 9°	9°-10°	10 °-11°	11°-12°	12°-13°
January	33.5613	33.8517	34.6933	33.07	35.934
February	34.9364	36.0257	37.1422	36.745	37.4382
March	35.795	38.618	38.0933	37.054	37.6816
April	35.605	38.054	41.29	38.96	40.158
May	35.97	36.81	43.65	38.89	42.26
June	32.83	34.03	40.1367	36.35	36.73
July	34.27	32.47	39.49	33.97	33.65
August	32.07	37.13	39.04	34.3833	31.13
September	34.082	36.51	35.058	35.3	32.37
October	33.7445	34.7433	35.1962	33.4233	34.59
November	34.44	34.5033	34.8833	34.95	34.55
December	34.1986	33.65	33.238	31.8718	34.24

Tab.3 highest monthly average of 2009 in Degree Celsius

On the other hand, unusually low temperatures in July (-3.79°C at 8°–9° latitude and -4.83°C at 12°–13° latitude) sharply contrasted with Kerala's tropical environment, most likely due to mistakes in the sensors or data processing. Despite irregular readings like 4.29°C in June (10°–11° latitude), valid monsoon lows, including 8.17°C in July (9°–10° latitude) and 8.19°C in June, matched the anticipated cooling from cloud cover. Fig.8 will be showing the lowest average of the year. Tab.4 showing the lowest monthly average of the year 2009.

2009 Highest Monthly Average LST	8°- 9°	9°-10°	10 °-11°	11°-12°	12°-13°
January	33.5613	33.8517	34.6933	33.07	35.934
February	34.9364	36.0257	37.1422	36.745	37.4382
March	35.795	38.618	38.0933	37.054	37.6816
April	35.605	38.054	41.29	38.96	40.158
May	35.97	36.81	43.65	38.89	42.26
June	32.83	34.03	40.1367	36.35	36.73
July	34.27	32.47	39.49	33.97	33.65
August	32.07	37.13	39.04	34.3833	31.13
September	34.082	36.51	35.058	35.3	32.37
October	33.7445	34.7433	35.1962	33.4233	34.59
November	34.44	34.5033	34.8833	34.95	34.55
December	34.1986	33.65	33.238	31.8718	34.24

Tab.4 lowest monthly average of 2009 in Degree Celsius.

Extreme pre-monsoon heat was observed seasonally from April to May, especially in central latitudes (10° – 13°), however data from the monsoon months (June–September) were inconsistent, with improbable lows overshadowing legitimate cooling tendencies. Stabilization was brought about by the post-monsoon season (October–December), as seen by the 34.59°C in October (12° – 13° latitude) and the modest NDVI improvements (0.878 at 11° – 12° latitude). Fig.5 showing the NDVI value for this year.

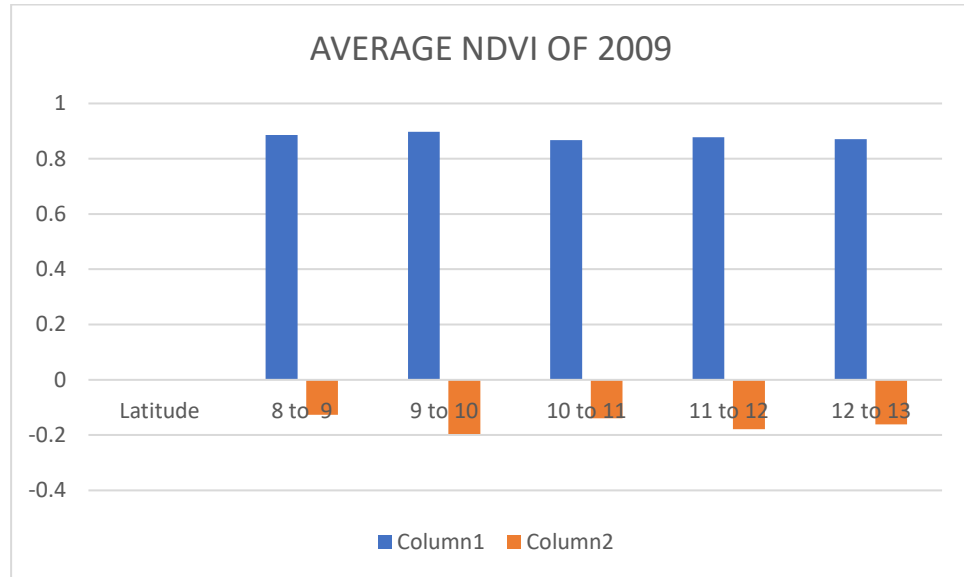
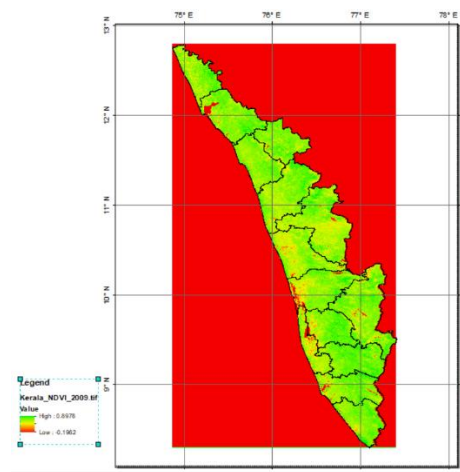


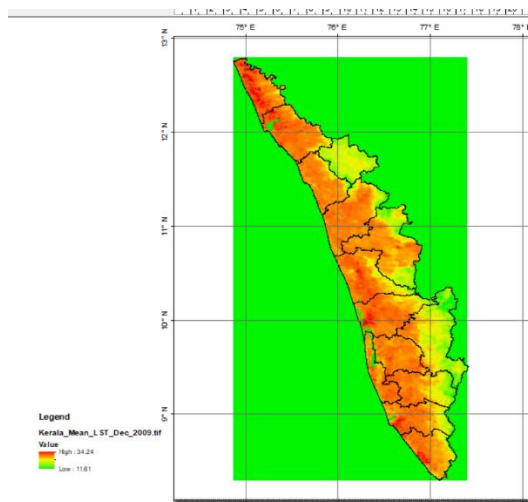
Fig.5 NDVI values for the year 2009

The NDVI-temperature relationship demonstrated the moderating effect of vegetation: low/negative NDVI areas (8° – 9° , 12° – 13° latitude) experienced increased variability, such as -3.79°C in July, probably as a result of urban or arid landscapes speeding up heat loss, while high-NDVI zones (9° – 10° latitude, NDVI: 0.8978) maintained milder temperatures (e.g., 36.03°C in February) through evapotranspiration.

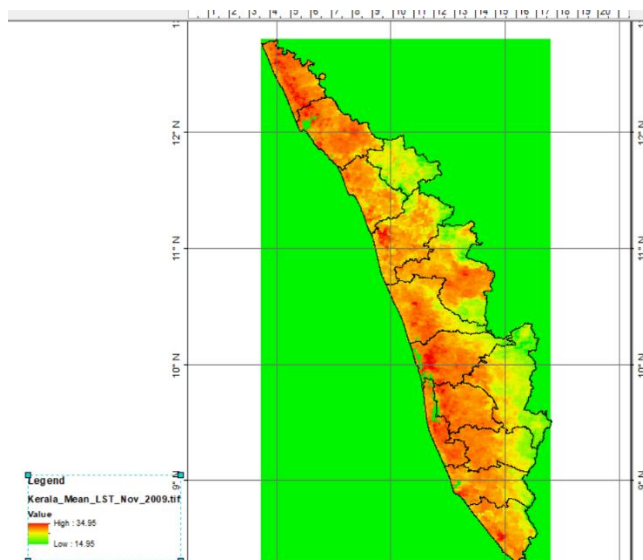
Here are some snapshots of LST analysis of 2009;



plt.5. NDVI Map of Kerala, 2009



plt.6 LST map of Kerala, December 2009



plt.7 LST map of Kerala, November 2009

4.1.3. 2014

An examination of Kerala's 2014 NDVI and temperature data shows a complicated interaction between vegetation, latitude, and seasonal dynamics that is characterized by significant anomalies and extremes. Although dense vegetation usually moderates heat, the 10–11° latitude had the greatest temperatures, reaching a peak of 42.04°C in March, along with the highest NDVI value of 0.9995. This anomaly implies that the cooling benefits of vegetation may have been offset by localized variables like decreased cloud cover, urbanization, or land-use patterns. On the other hand, the 12–13° latitude showed consistently high temperatures (such as 39.88°C in April) but a lower NDVI (0.9487), which is consistent with the idea that heat retention occurs in

areas with less vegetation. However, there were discrepancies at the lowest temperatures. Tab.5 showing the highest monthly temperature of 2014.

2014 Monthly LST	Highest Average	8°- 9°	9°- 10°	10°- 11°	11°- 12°	12°- 13°
January		34.5775	33.45	34.6309	32.4567	34.2976
February		35.358	36.0445	36.9013	35.7356	37.1371
March		38.1786	39.0067	42.041	40.0226	39.879
April		35.154	40.29	41.996	38.985	39.89896
May		36.19	37.3327	39.77	35.0567	35.73
June		36.97	34.467	39.6775	33.7129	35.71
July		32.09	33.45	36.99	35.75233	35.21
August		34.31	35.63	36.71	37.2233	33.89
September		35.38	35.45	39.69	35.9725	33.19
October		37.41	35.03	35.89	34.95	32.74
November		35.175	34.24	34.6633	32.8731	34.2809
December		32.9411	32.8167	33.1645	31.7755	33.3136

Tab.5 highest monthly average of 2014 in Degree Celsius.

While true lows (such as 19.22°C in January) coincided with modest NDVI (0.9071), indicating vegetation's ability to reduce extremes, the 9–10° latitude recorded an unrealistic -6.13°C in September—is a data mistake. Data dependability difficulties were further highlighted by the doubtful -2.05°C recorded in June at the 10–11° latitude. Tab.6 showcase the lowest monthly average temperature of the year 2014.

2014 Lowest Monthly Average LST	8°- 9°	9°-10 °	10°- 11°	11°- 12°	12°- 13°
January	20.3033	19.2247	17.7119	19.2535	24.2936
February	22.9957	21.675	20.5713	21.77	25.9607
March	20.3033	22.3033	22.1224	23.4373	28.9607
April	25.225	24.3407	21.1733	21.118	27.47
May	19.63	21.3078	19.84	17.55	26.1186
June	17.03	7.55	-2.05	9.53	11.81
July	11.59	15.69	5.37	13.43	9.65
August	18.08	15.31	-1.61	10.99	9.51
September	17.53	-6.13	10.55	15.682	20.63
October	20.17	9.37	6.63	10.67	23.14
November	21.56533	10.89	16.15	17.3	23.7009
December	19.1329	20.4843	16.2767	17.83	23.9823

Tab.6 lowest monthly temperature of 2014 in Degree Celsius.

Due to pre-monsoon solar intensity, March through May was the hottest season overall, with consistent highs across all latitudes. While legitimate entries (such as 15–20°C) matched with normal monsoon cooling, the data from the southwest monsoon (June–September) was inconsistent and included irrational sub-zero lows. Kerala's tropical environment was reflected in the stable temperatures between 20 and 35°C during the post-monsoon season (October to December). Higher NDVI zones (e.g., 10–11° latitude) somewhat moderated daytime heat (e.g., 34.63°C in January vs. 34.57°C in 8–9° latitude), but failed to prevent extremes, probably because of competing factors like urbanization or solar radiation. This highlighted the complex role of vegetation. Lower NDVI regions (12–13° latitude) showed greater temperatures, highlighting the significance of less vegetation in absorbing heat. Fig.12 shows the NDVI value of corresponding value.

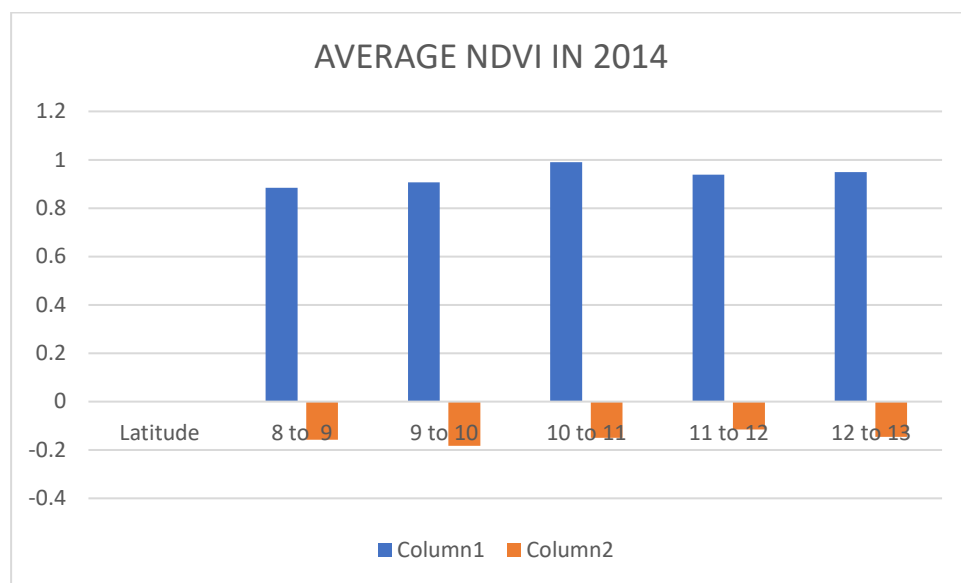
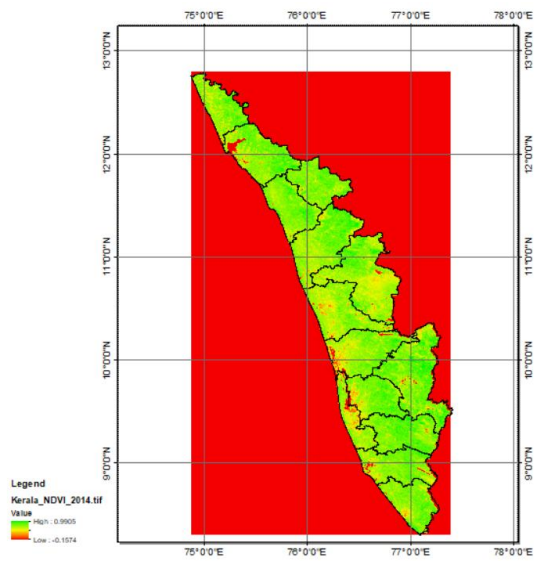


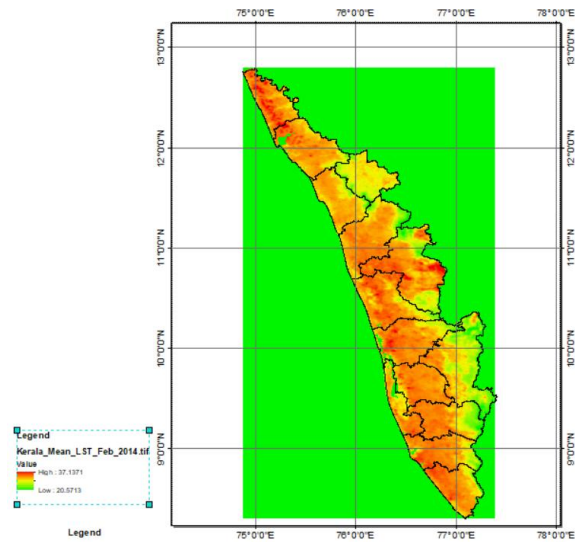
Fig.6. NDVI of Kerala 2014

The climate of Kerala is drastically at odds with critical data anomalies like -6.13°C in September and -2.05°C in June, which are probably caused by sensor failures. Especially during the monsoon season, these discrepancies make it difficult to see obvious patterns. The need for better quality control in climatic assessments is highlighted by the fact that, despite vegetation (NDVI) somewhat reducing temperatures, seasonal solar patterns, local environmental factors, and data inconsistencies dominated Kerala's thermal profile in 2014.

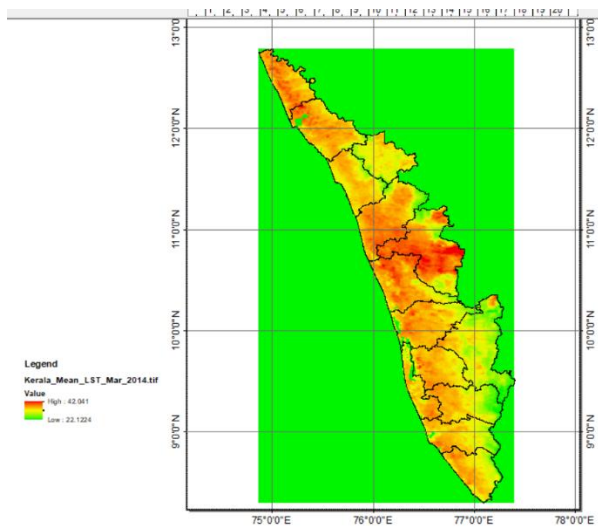
Here are some snapshot of LST analysis of 2014;



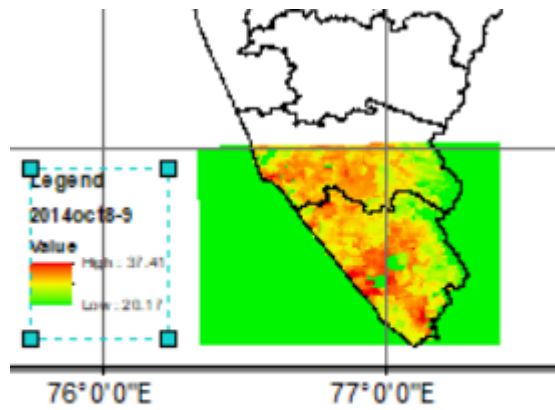
plt.8. NDVI map of Kerala,2014



plt.9 LST map of Kerala, February 2014



plt.10 LST map of Kerala, March 2014



plt.11 Clip image of LST map of 8°-9° latitude ,of 2014.

4.1.4. 2019

The 2019 study of Kerala's temperature and NDVI (Normalized Difference Vegetation Index) revealed notable seasonal and regional fluctuations caused by vegetation cover, latitude-based zones, and anomalies in the data. Distinct thermal trends were discovered by classifying by latitude classes. The most extreme temperatures were found in the 10–11° latitude zone, with record highs of 42.95°C in June and 41.35°C in April. These highs were probably caused by urban heat island effects or decreased vegetation density (NDVI 0.8938). Although the 8–9° latitude also had high temperatures (37.41°C in February, for example), the extremes were not as great. Tab.7 showcasing the highest monthly average LST of 2019.

2019 Highest Average Monthly LST	8° - 9°	9° - 10°	10° - 11°	11° - 12°	12° 13°
January	34.6111	33.6247	34.1395	33.6017	34.4909
February	37.4138	35.0553	37.327	36.956	37.3656
March	39.33	39.4607	40.2582	38.8595	37.7039
April	38.138	39.7367	41.35	41.35	38.6591
May	35.7878	37.13	40.4	37.7567	40.13
June	38.11	33.87	42.95	36.29	35.57
July	36.07	34.09	39.79	37.65	34.21
August	33.75	34.61	34.85	38.53	32.59
September	36.79	34.25	39.15	37.37	33.19
October	32.59	35.07	38.65	35.12	36.11
November	36.69	33.93	34.7671	33.3567	32.566
December	35.285	33.8814	34.454	32.7329	32.8333

Tab.7 highest monthly average of 2019 in Degree Celsius

Ironically, low temperatures of 0.29°C in June (perhaps a data error) and more reasonable lows of 17.43°C in December were also recorded in the 10–11° latitude. Further indicating data inconsistencies, the 11–12° latitude recorded extreme monsoon-season lows (e.g., 1.47°C in July), which contrasted with its comparatively high NDVI (0.8993). July temperatures, for example, varied significantly with latitude: 1.69°C (9–10°), 1.47°C (11–12°), and 3.66°C (8–9°), highlighting the necessity for thorough data validation. Fig.14 shows the lowest monthly average of the year 2019.

2019 Lowest Average Monthly LST	8°- 9°	9°- 10°	10°-11°	11°-12°	12°-13°
January	20.8194	20.5714	18.7573	20.5733	24.965
February	22.3117	21.2936	20.3932	21.771	26.9622
March	25.0967	23.9786	22.7573	23.639	27.8167
April	25.2929	25.1265	23.6678	23.6678	27.9229
May	21.92	22.61	21.6479	21.38	25.474
June	15.61	5.35	0.29	10.47	1.97
July	3.65	1.69	6.43	1.47	11.45
August	10.51	17.29	10.39	13.29	18.61
September	13.99	7.71	10.05	6.53	17.37
October	15.49	12.91	15.11	6.69	18.27
November	17.72	16.94	17.3871	18.736	24.2962
December	19.9033	19.3767	17.43	18.4654	24.3543

Tab.8 lowest monthly average temperature of 2019 in Degree Celsius

Significant variety was observed in both monthly and seasonal trends. While the summer months of March through May had continuously high averages (e.g., 40.25°C in March), the hottest months were April through June, with a peak temperature of 42.95°C in June (10–11° latitude). The chilly months of June through September, on the other hand, showed unpredictable lows, with unlikely readings such as 0.29°C in June and 1.47°C in July, which clash with Kerala's tropical environment. Valid monsoon-season lows (10–20°C) matched the usual cooling effects. December saw steady lows between 17 and 24°C, which was indicative of post-monsoon moderation. The southwest monsoon (June–September) had erratic lows that probably concealed milder actual temperatures (15–25°C), whereas the summer (March–May) was the hottest season. Fig.15 shows the NDVI values of Kerala for 2019.

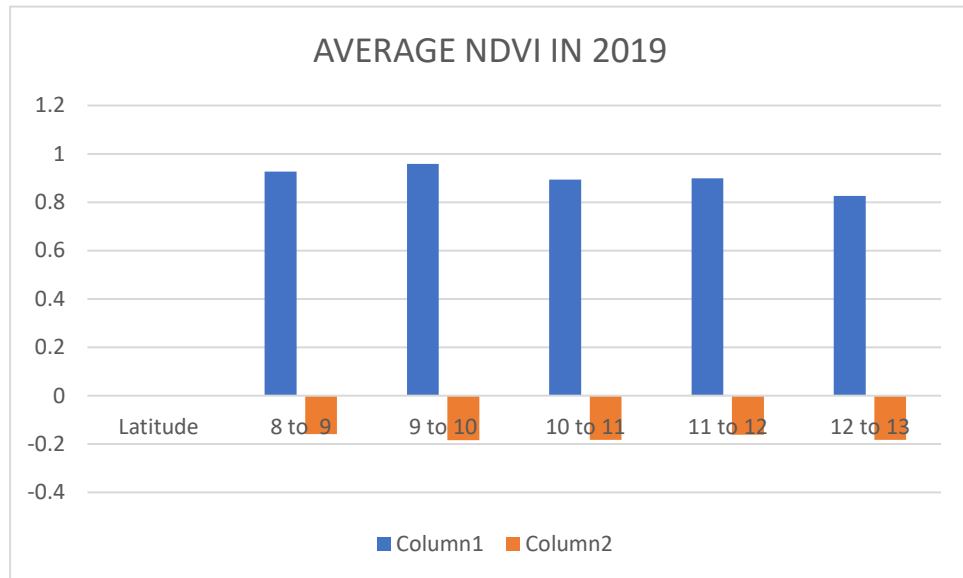
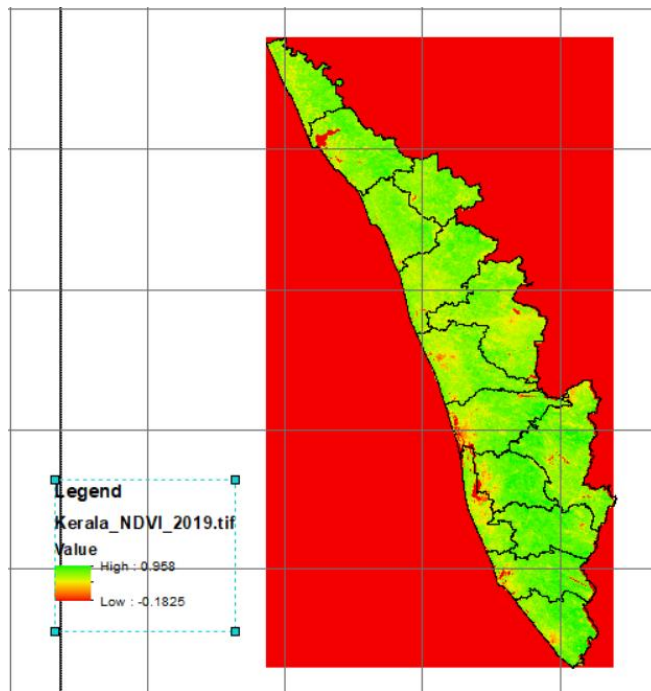


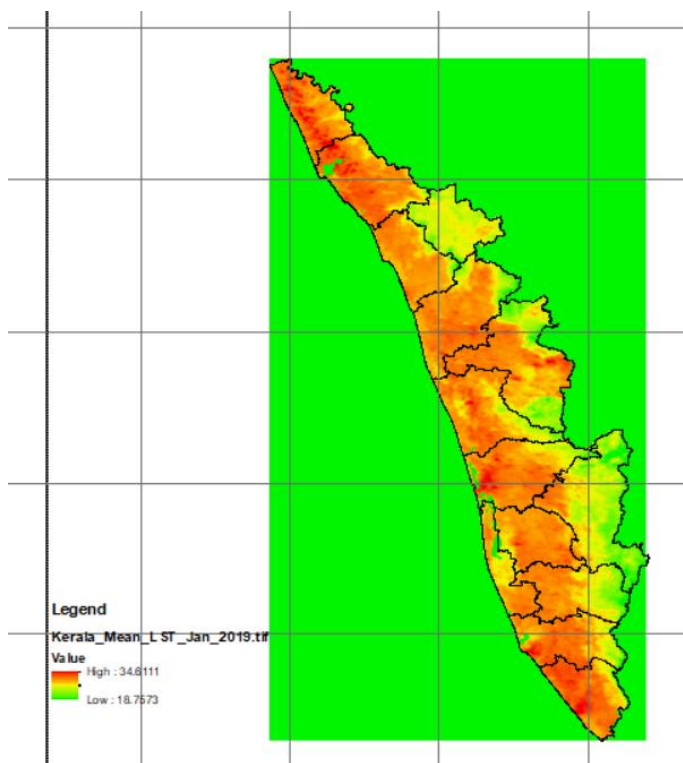
Fig.7. NDVI value of Kerala for 2019

The link between temperature and NDVI showed subtle relationships. Greater NDVI regions, like the 9–10° latitude (NDVI 0.958), had moderate temperatures (e.g., 35.05°C in February), indicating the significance of vegetation in mitigating heat. Nonetheless, inconsistent extremes, including unusual lows during monsoons, were observed at the 11–12° latitude (NDVI 0.8993), suggesting problems with data dependability. Because sparse vegetation has a limited ability to buffer temperature extremes, regions with lower NDVI, such as the 12–13° latitude (NDVI 0.8257), showed gentler highs (38.65°C in April) and lows (1.97°C in June). Notably, the extreme highs in the 10–11° latitude range, even with a moderate NDVI, suggested that thermal dynamics were dominated by non-vegetation variables like urbanization or changes in land use. In summary, the coldest latitudes were 10–11° (post-validation) and 12–13° (mild highs associated with low NDVI), whereas the warmest were 10–11° (extreme highs) and 11–12° (summer highs). Summer (March–May) was associated with seasonal peaks, but monsoon data needed closer examination. The moderating influence of NDVI was noticeable in areas with a lot of vegetation, but in urbanized latitudes, human factors took centre stage.

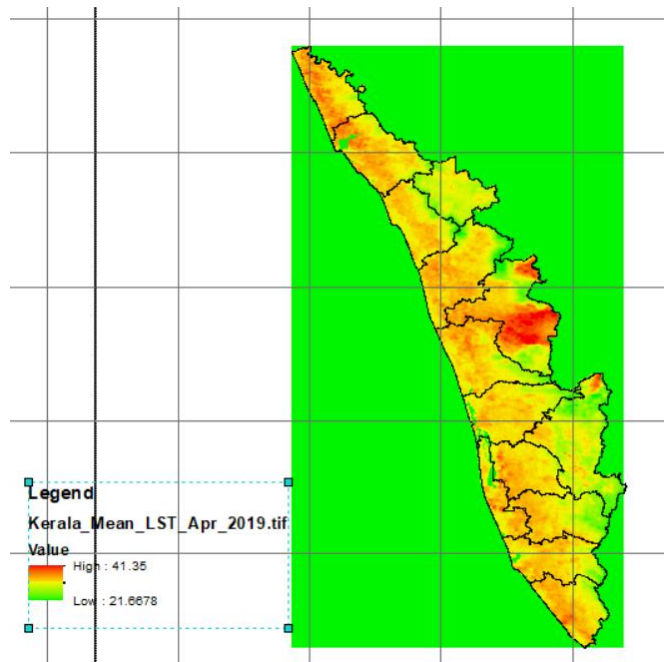
Here are some snapshots of the LST analysis of 2019;



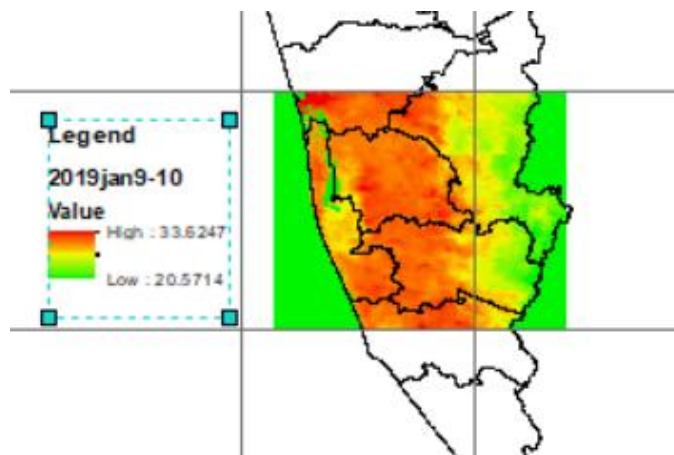
plt.12 NDVI map of Kerala , 2019



plt.13 LST map of Kerala, January 2019



plt.14 LST map of Kerala, April 2019



plt.15 Clip Image of 9°-10° latitude of the LST map of the year 2019

4.1.5. 2024

Kerala's 2024 temperature and NDVI (Normalized Difference Vegetation Index) research highlights the intricate relationship between elevation, vegetation, and data reliability by displaying remarkable thermal patterns and anomalies across altitude classes. Extreme temperature differences are highlighted by classifying by altitude classes. Despite its very high NDVI (0.9823), the 10–11° altitude zone recorded record highs, reaching a peak of 46.13°C in May and 40.75°C in April. The traditional cooling effects of vegetation are contradicted by such excessive heat in a densely vegetated zone, suggesting that the data may be inaccurate or that non-vegetation factors such as urbanization, isolated deforestation, or sensor malfunctions may be at play. With modest NDVI (0.9425), the 11–12° altitude also saw high temperatures

(e.g., 39.07°C in April), indicating heat retention in less vegetated areas. Tab.9 shows the highest monthly averages of the year 2024.

2024 Highest Monthly Average LST	8°- 9°	9°- 10°	10°- 11 °	11°- 12°	12°- 13°
January	34.4485	32.4571	32.5007	31.4871	31.4686
February	35.4844	34.6738	35.5929	33.448	34.7356
March	36.249	35.1627	37.056	35.5658	35.8416
April	37.26	37.0725	40.7575	39.0793	37.9188
May	35.09	36.41	46.13	38.5233	37.55
June	34.43	35.25	34.07	38.41	31.37
July	33.47	31.55	37.09	29.77	28.67
August	36.69	33.67	32.77	32.43	34.43
September	35.27	34.05	35.254	33.42	35.27
October	35.45	34.09	34.75	31.8775	32.21
November	34.71	32.294	33.0633	31.3753	30.3513
December	32.964	30.9433	32.5257	30.3632	29.9377

Tab.9 highest monthly average of 2024 in Degree Celsius.

The 8–9° altitude zone, on the other hand, recorded ridiculous extremes, like - 3.87°C in August, which is not consistent with Kerala's climate. This was probably the result of sensor error. Its mild NDVI (0.9123) corresponds with valid lows in this zone, such as 17.40°C in January, which represent normal vegetation-mediated cooling. Questions about data trustworthiness were further highlighted by the 9–10° altitude zone, which also displayed dubious extremes, such as 3.75°C in July. Tab.10 shows the lowest average monthly LST in 2024.

2024 Lowest Monthly Average LST	8°- 9°	9°-10°	10°- 11°	11°- 12°	12°-13°
January	20.6867	20.1714	17.4025	19.5735	23.8973
February	22.1685	21.025	19.2067	21.0517	25.619
March	24.565	24.1309	22.2152	22.7012	26.819
April	24.8833	22.45	22.3775	23.1517	26.784
May	21.73	10.37	14.73	8.57	24.7367
June	14.67	7.85	7.75	5.39	19.31
July	18.63	3.75	7.37	7.91	18.19
August	-3.87	16.25	7.37	10.36	19.87
September	18.12	11.81	13.0329	17.27	18.12
October	18.91	15.43	15.92	18.13	18.41
November	21.265	13.37	13.37	15.13	23.155
December	19.6171	19.736	17.818	17.5133	22.91

Tab.10. lowest monthly average of 2024 in Degree Celsius.

Kerala's seasonal temperature trends are influenced by anomalous data, but they generally follow the tropical monsoon climate. With severe highs (such as 46.13°C in May) centred in the 10–11° altitude range, April–May were the hottest months. The March–June period saw high temperatures that were consistent with pre-monsoon heat, averaging 37–38°C. On the other hand, the coldest months (June through August) displayed unpredictable lows, notably August's physiologically implausible -3.87°C. This time frame's acceptable lows (such as 17–19°C in December) correspond with monsoon cooling. Based on seasonal research, the hottest season is summer (March–May), which is caused by pre-monsoon aridity. If anomalies are taken out of the equation, the southwestern monsoon (June–September) is expected to bring gentler temperatures (15–25°C). Gradual cooling was seen throughout the post-monsoon season (October–December). Fig.18 shows the NDVI value of Kerala 2024.

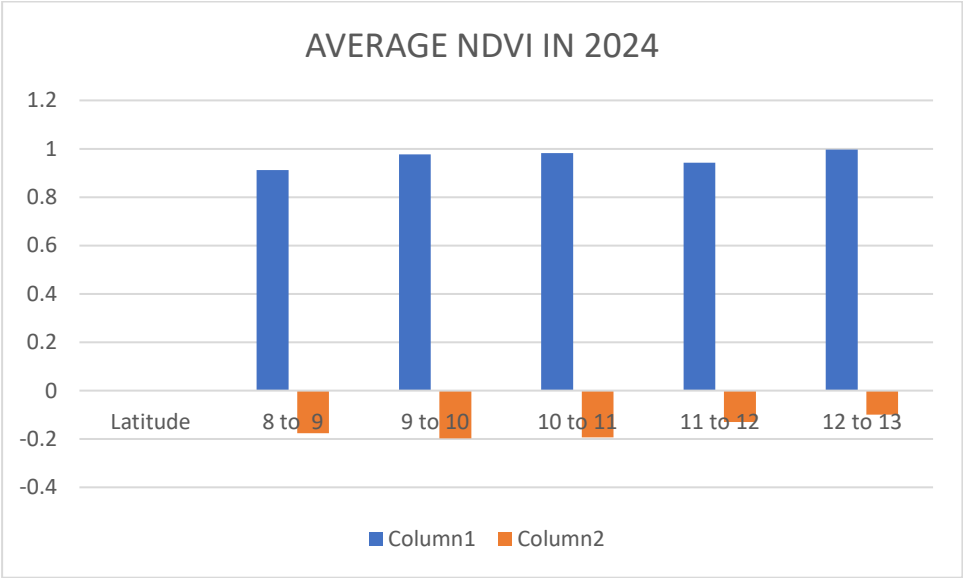
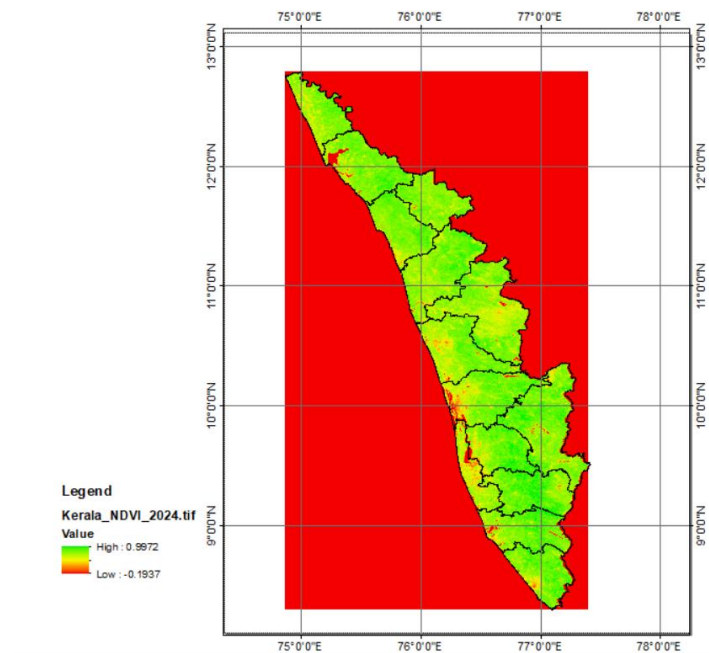


Fig.8. NDVI of Kerala 2024

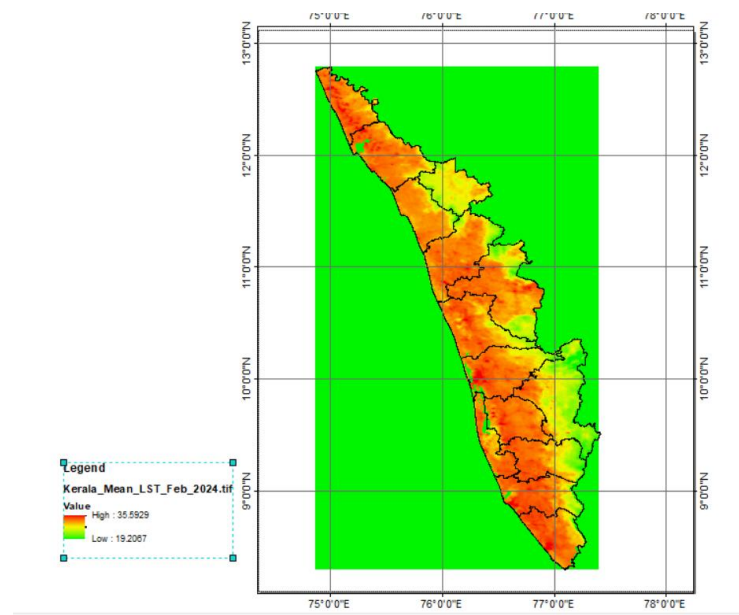
The connection between temperature and NDVI emphasizes how vegetation influences climate. The cooling effect of vegetation was evident in zones with greater NDVI, such as the 12–13° altitude zone (NDVI: 0.9972), which showed moderate temperatures (e.g., 37.55°C in May). This pattern was further supported by mild highs (such as 34.67°C in February) observed at the 9–10° altitude (NDVI: 0.9768). Higher temperature extremes (such as 36.69°C in August) were observed in lower NDVI zones, such as the 8–9° altitude (NDVI: 0.9123), indicating a decreased capability for cooling because of the scant vegetation. Anomalies in the 10–11° altitude range, where intense heat and high NDVI coexist, however, suggest that confounding variables like urban heat islands or inaccurate sensors may be offsetting the effects of vegetation.

The 2024 temperature patterns in Kerala highlight the **10–11° altitude zone** as the hottest due to extreme summer highs, while the **12–13° altitude** (with high NDVI) and **9–10° altitude** (with valid lows) emerge as cooler regions. Seasonal trends align with expected monsoonal influences, though anomalous data obscure monsoon-month interpretations. NDVI generally correlates with temperature moderation, but outliers suggest non-vegetation factors dominate in certain zones.

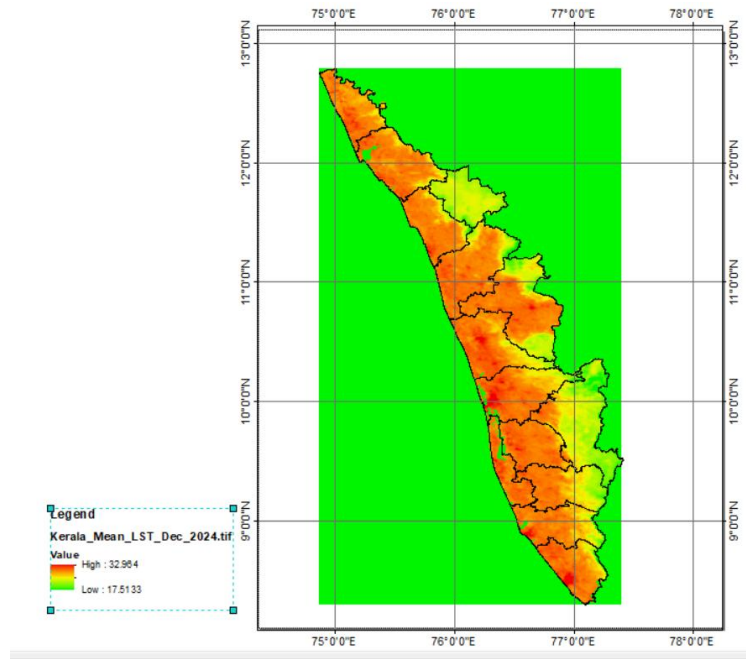
Here are some snapshots of LST analysis of 2024;



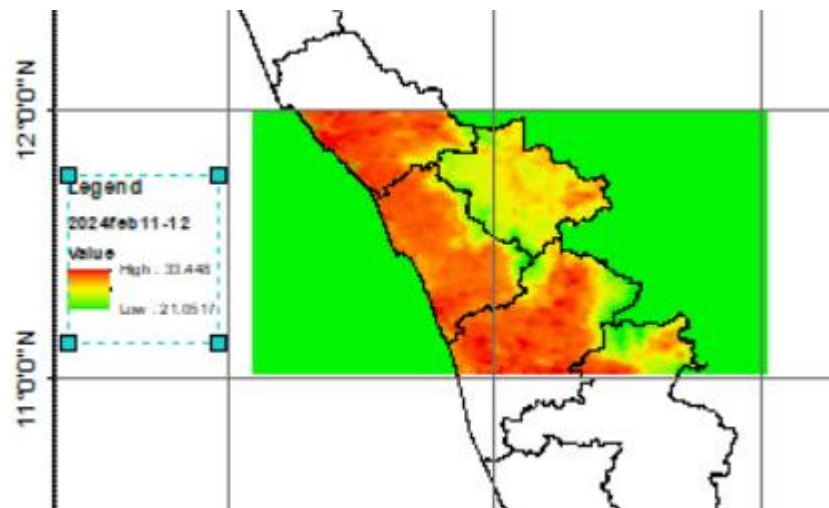
plt.16 NDVI map of Kerala, 2024



plt.17 LST map of Kerala, February 2024



plt.18 LST map of Kerala, December 2024



plt.19- Clip image of LST map, 11°-12° latitude of 2024

4.2. ALTITUDINAL WISE

4.2.1. 2004

Significant differences existed in temperature extremes between altitude zones. Lower elevations (CLASS 1–2: 0–100 meters) consistently had the greatest monthly average maxima. CLASS 2 (0–100 meters) predominated in the summer (March–May), with lowland areas experiencing highs of 42.38°C in March and 41.80°C in April due to strong solar radiation. Significant summer heat was also experienced by CLASS 3 (100–500 meters), which recorded the highest temperature of all classes in March at 43.17°C. Mid-altitudes (CLASS

4: 500–1000 meters) observed anomalous monsoon-season highs, including 37.07°C in August, which were probably caused by heat-trapping clouds. Tab.11 shows the highest monthly average altitude wise for 2004.

2004 Altitude Wise Average Maximum LST	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY	31.49086	35.48333	34.99117	32.64888	32.10159	27.61091
FEBRUARY	33.03632	38.265	39.06158	34.513	20.79095	29.79095
MARCH	34.50692	42.37663	43.16625	41.857	39.34111	35.55095
APRIL	34.79	41.79857	41.24538	38.07154	34.13818	31.87706
MAY	31.97	38.15	38.73	34.27	30.7966	29.99
JUNE	29.23	34.31	31.37	31.51	28.69	22.322
JULY	31.09	34.31	33.67	36.81	31.45	23.85
AUGUST	28.525	32.84333	33.38	37.065	31.4071	24.05
SEPTEMBER	32.5166	37.41	35.97	39.314	30.24	23.93
OCTOBER	31.33286	36.518	33.38666	32.845	29.46	26.59
NOVEMBER	30.246	33.70538	33.33923	29.03571	27.2975	23.53571
DECEMBER	30.63125	34.30083	34.3275	30.03316	28.99666	26.46636

Tab.11.highest monthly average altitude wise of 2004 in Degree Celsius.

The coldest minimum temperatures were seen at higher elevations (CLASS 5–6: 1000–3000 meters). CLASS 5 (1000–2000 meters) recorded unusual lows, such as 9.57°C in May and 3.8°C in November, which are in contrast to Kerala's tropical climate norms, while CLASS 6 (2000–3000 meters) had a minimum of 9.42°C in July. CLASS 1 (>0 meters) and lower altitudes maintained the warmest minima throughout the year (e.g., 30.23°C in March). In CLASS 2 (0–100 meters), a significant anomaly was noted when the minimum temperature in August fell to 0.43°C, which is not consistent with local climate conditions and is most likely the result of sensor mistake. Tab.12 showing the lowest monthly averages of corresponding year in altitudinal wise.

2004 Altitude Wise Average Minimum LST	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY	26.03846	25.77	24.11087	22.102	18.7387	18.72739
FEBRUARY	28.11762	25.675	25.99	23.916	20.79095	19.96826
MARCH	30.2333	28.1033	28.33842	25.75091	23.5766	23.49526
APRIL	28.574	27.665	27.0788	24.89833	21.47923	20.89706
MAY	23.93	17.01	12.36	17.53	9.57	14.69
JUNE	23.2966	9.99	12.55	12.27	8.39	11.76333
JULY	20.69	14.97	16.37	14.05	7.61	9.42
AUGUST	22.88	0.43	19.13	8.99	11.15	11.15
SEPTEMBER	25.77	7.07	15.29	17.05	11.0566	9.87
OCTOBER	25.01	22.8966	21.53	19.9625	17.535	18.15
NOVEMBER	26.616	22.01	15.85	14.77	3.8	16.49
DECEMBER	26.19761	23.79	23.4361	21.1947	19.13571	19.14429

Tab.12. lowest monthly average altitudinal wise of 2004 in Degree Celsius.

Seasonal temperature trends aligned broadly with Kerala's tropical monsoon climate but included localized deviations:

- **Summer (March–May):** The highest temperatures, which peaked in March at 43.17°C (CLASS 3), were mostly found at lower elevations (CLASS 2–3: 0–500 meters). Because of lowland heat retention, minimum temperatures in CLASS 1 (>0 meters) continued to be the highest. \
- **Monsoon (June–September):** Maximum temperatures at mid-altitudes (CLASS 4: 500–1000 meters) showed surprising peaks (e.g., 39.31°C in September), which may have been caused by less cooling at night under continuous cloud cover. The warmest minimum temperature Monsoon (June–September): Mid-altitudes (CLASS 4: 500–1000 meters) showed unforeseen The warmest minimum temperatures (e.g., 25.77°C in September) were maintained by CLASS 1 (>0 metres).
- **Winter (October–February):** Warmth was again strongest at lower elevations, with CLASS 2 (0–100 meters) having the highest winter maximum (38.27°C in February) and CLASS 1 (>0 meters) recording the warmest minima (28.12°C in February).
- The research confirmed the altitude-temperature relationship, which followed the environmental lapse rate (~6.5°C per KM) and showed that both maxima and minima declined with height. The monsoon highs of CLASS 4 and the severe lows of CLASS 5 in May and November, however, were observed as exceptions, indicating either data errors or microclimatic factors (such as valley inversions). The 0.43°C in August (CLASS 2) and the 9.57°C in May (CLASS 5) were two examples of anomalies that further highlighted possible sensor issues or recording problems.

4.2.2. 2009

In 2009, the extremes of temperature varied significantly with height. The highest monthly average maxima were the most common at lower elevations (CLASS 1–3: 0–500 meters). Due to strong pre-monsoon solar heating, CLASS 3 (100–500 meters) recorded the most extreme readings, reaching a peak of 41.29°C in April and 43.65°C in May, the year's maximum temperature. Unusual monsoon anomalies were also seen in this zone, as June temperatures defied normal monsoon cooling patterns by rising to 38.01°C. Following closely behind was CLASS 2 (0–100 meters), which showed high temperatures (37.14°C in February) indicative of lowland heat retention. The monsoon season

(June–September) had the highest temperatures in mid-altitudes (CLASS 4: 500–1000 meters), with July recording 37.57°C. This pattern points to heat-trapping effects caused by persistent cloud cover, which has been seen in previous years. Tab.13 showing the highest maxima in the elevation classes in 2009.

2009 Altitude Aise average Maximum LST	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY	31.9947	35.89381	35.934	32.317	30.21364	27.3823
FEBRUARY	36.0257	37.14364	37.43818	36.745	33.63304	30.40083
MARCH	38.6179	38.0933	38.0655	37.054	34.62052	31.644
APRIL	37.27	40.955	41.29	38.22571	36.46733	30.4
MAY	33.95	42.39	43.65	38.042	34.916	30.285
JUNE	30.75	36.73	38.01	37.333	32.81	28.738
JULY	30.15	34.09	31.95	37.57	32.29	23.21
AUGUEST	29.84	37.13	34.21	35.1033	30.702	23.91
SEPTEMBER	29.96333	36.51	33.45	35.08	29.446	23.99666
OCTOBER	31.43	34.6566	34.59	33.39154	29.8066	24.0766
NOVEMBER	31.72	34.95	34.0075	30.034	28.698	24.61285
DECEMBER	31.79545	34.24	33.3857	29.07728	27.48555	24.88429

Tab.13. highest monthly average in elevation classes on 2009 in Degree Celsius

The coldest minimum temperatures were seen at higher elevations (CLASS 5–6: 1000–3000 meters). In July, CLASS 6 (2000–3000 meters) recorded a lowest temperature of 8.95°C, which is consistent with the predicted cooling at high altitudes. In contrast, CLASS 5 (1000–2000 meters) recorded an unrealistic low of -3.79°C, which is a glaring anomaly that is not consistent with Kerala's tropical climate and most likely the result of a sensor error. CLASS 1 (>0 meters) and lower altitudes maintained the hottest minimum temperatures throughout the year, with January temperatures reaching 26.71°C. Another anomaly was noted in CLASS 4 (500–1000 meters), when the minimum temperature in July dropped to -3.79°C, which is also an invalid value for the area. Tab.14 showing the average minimum temperatures on 2009.

2009 Altitude Wise Average Minimum LST	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY	26.71087	26.67782	23.2453	20.67923	18.63182	19.10636
FEBRUARY	27.25909	26.63	26.90727	24.16	20.79	20.77833
MARCH	29.3175	27.9233	27.648	25.13941	21.6366	22.506
APRIL	27.62714	24.06	26.12272	24.3825	21.97851	22.25615
MAY	26.78333	22.95	24.15	22.565	18.89	20.4125
JUNE	23.81	8.19	15.43	4.29	10.39	14.25
JULY	18.46	8.17	17.05	-3.7899	-3.7899	8.95
AUGUEST	24.05	15.53	9.05	8.61	10.6	8.37
SEPTEMBER	24.47	10.67	20.39	9.85	14.59	12.61
OCTOBER	28.4	25.93	24.136	21.87666	17.93	17.75181

NOVEMBER	26.8388	20.43	21.27	18	14.95	18.01333
DECEMBER	26.415	11.61	23.11666	20.52	17.35	17.24777

Tab.14. lowest monthly average LST of elevation classes in Degree Celsius.

Seasonal Variation

Seasonal temperature trends in 2009 largely mirrored Kerala's tropical monsoon climate but included notable deviations:

- **Summer (March–May):** Maximum temperatures were mostly seen at lower elevations; in May, CLASS 3 (100–500 meters) peaked at 43.65°C. With a temperature of 40.96°C in April, CLASS 2 (0–100 meters) followed, reflecting strong solar radiation in the lowlands. With an average of 26.78°C in May, the lowest temperatures were still highest in CLASS 1 (>0 meters).
- **Monsoon (June–September):** The July high for mid-altitude CLASS 4 (500–1000 meters) was 37.57°C, most likely as a result of less cooling at night under heavy cloud cover. In contrast, CLASS 3 recorded an unusual June high of 38.01°C, which defied normal monsoon cooling. Though the -3.79°C estimate in CLASS 5 is still debatable, higher altitudes (CLASS 5–6) saw a substantial cooling
- **Winter (October–February):** The winter months of October through February saw consistent warmth at lower elevations once more. In February, CLASS 2 (0–100 meters) reached a peak of 37.14°C, while CLASS 1 (>0 meters) maintained the warmest minimum temperatures (26.42°C) in December.

Since both maxima and minima typically decreased with height, the research confirmed the altitude-temperature link. In comparison to CLASS 1 readings, CLASS 6 minima were 5–10°C colder. A few anomalies did, however, stand out, such as the monsoon peaks of CLASS 4 and the exceptionally high June maximum of CLASS 3, which may indicate data errors or microclimatic factors (such as localized cloud dynamics or valley heat retention). Anomalies such as the -3.79°C in July for CLASS 4 and CLASS 5 and the 38.01°C in June for CLASS 3 further highlighted possible sensor issues or recording problems.

According to the 2009 thermal analysis, Kerala's hottest regions are found at lower elevations (CLASS 2–3), when summer temperatures can reach 43°C. Although abnormal lows (-3.79°C) probably reflect data mistakes, higher elevations (CLASS 5–6) continued to be the coldest. Seasonal changes showed that lowlands experienced summer heat dominance, mid-altitude monsoon fluctuation, and winter lowland warmth that was constant. The dataset's usefulness for planning climate resilience and agricultural adaptation in Kerala's varied terrain will be increased by resolving data errors and investigating microclimatic factors.

4.2.3. 2014

Kerala showed distinct patterns in temperature extremes across its elevation classes, which ranged from CLASS 1 (>0 meters) to CLASS 6 (2000–3000 meters):

Maximum Temperatures: Because of the strong solar heating in lowland and coastal regions, the lowest elevations (CLASS 2: 0–100 meters) dominated most months, with maximum of 42.04°C in March and 39.77°C in May. CLASS 4 (500–1000 meters) dominated during monsoon months (e.g., 37.98°C in June and 38.57°C in September), whereas CLASS 3 (100–500 meters) had an exception, recording 41.99°C in April. These peaks at mid-elevation point to heat retention caused by monsoon clouds. The coldest maximum, such as 24.94°C in January and 24.49°C in December, were consistently recorded at the highest elevations (CLASS 6: 2000–3000 meters), following the usual altitude-driven cooling pattern. Tab.15 shows the highest monthly averages in the year of 2014.

2014 wise average maximum LST	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY	31.9725	34.63087	34.0271	30.2322	28.338	24.93571
FEBRUARY	34.19875	36.90125	37.13706	34.67632	32.55588	28.87941
MARCH	37.75625	42.041	41.954	39.69551	35.17947	30.0988
APRIL	35.1233	41.555	41.996	40.28176	36.64666	29.639
MAY	38.63	39.77	39.52429	35.58571	32.89714	27.698
JUNE	30.1633	37.29	36.03	37.9825	34.0375	24.9033
JULY	30.65	37.21	33.45	36.16333	32.53	25.56
AUGUST	30.89	36.71	34.81	35.81	31.61	24.85
SEPTEMBER	32.09	35.45	33.9933	38.5725	32.045	24.05
OCTOBER	32.83	37.41	34.95	30.96	29.33	24.0433
NOVEMBER	31.615	35.175	33.286	30.3633	27.2588	23.35857
DECEMBER	31.57364	33.31364	32.85182	28.6957	27.6033	24.494

Tab.15. highest monthly average LST on 2014 in Degree Celsius.

The warmest minimum temperatures were consistently maintained by CLASS 1 (>0 meters), as evidenced by lowland heat retention, with 26.07°C in January and 30.01°C in March. The coldest minimums, on the other hand, were found at the highest elevations (CLASS 5–6), with CLASS 5 (1000–2000 meters) dipping to 17.55°C in May and CLASS 6 to 16.28°C in December. Incredibly low temperatures in CLASS 4 (-2.05°C in June), CLASS

2 (-1.61°C in August), and CLASS 5 (-6.13°C in September) were among the startling anomalies that surfaced during the monsoon. These readings were not compatible with Kerala's tropical climate and were probably caused by sensor mistakes or localized microclimates. Tab.16 shows the lowest monthly average LST for 2014.

2014 Wise Minimum LST	Altitude Average						
		CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY		26.06545	26.0558	23.99714	20.77625	18.06125	17.7119
FEBRUARY		28.08158	27.57632	26.28647	23.57133	20.8123	20.57125
MARCH		30.01	28.71	27.05296	24.85666	22.3033	22.1298
APRIL		29.66599	27.21	27.46692	24.84333	21.11799	21.17333
MAY		27.7833	19.63	22.14	20.66	17.55	19.84
JUNE		24.48	12.67	10.37	-2.0499	9.53	9.53
JULY		23.69	5.37	11.59	16.03	13.05	13.35
AUGUST		23.03666	-1.6099	6.21	15.19	4.63	8.57
SEPTEMBER		25.7433	14.77	15.93	7.65	-6.1299	16.245
OCTOBER		26.71	15.86	16.52	11.27	6.63	11.73
NOVEMBER		26.78	23.24	21.5322	18.3666	10.89	16.15
DECEMBER		25.9511	25.25	22.078	20.51142	17.23	16.27666

Tab.16. lowest monthly average LST on 2014 in Degree Celsius.

Seasonal Variations

Seasonal shifts further underscored the interplay between elevation and climate:

- **Winter (December–February):**

Class 2 reached its highest temperature of 36.90°C in February, while Class 6 recorded the lowest temperature (e.g., 24.94°C in January). Lower elevations continued to be the hottest during this time. Class 6 had the coldest minimum temperatures (e.g., 17.71°C in January) while Class 1 had the highest (e.g., 25.95°C in December).

- **Summer (March–May):**

CLASS 2–3 experiences the most heat throughout the summer (March–May), with CLASS 2 recording the highest summer maximum of 42.04°C in March. Higher elevations saw a modest drop in minimum temperatures, with CLASS 5 dropping to 17.55°C in May.

- **Monsoon (June–September):**

Unexpected maximums were seen at mid-elevations (CLASS 4), including 38.57°C in September, which was probably caused by heat that was trapped beneath continuous cloud cover. Maximum temperatures in July and August were dominated by lower elevations (CLASS 2), whereas CLASS 2 (-1.61°C in August) and CLASS 5 (-

6.13°C in September) had sharp, unusual decreases. These extremes show hyper-localized cooling in high-altitude valleys or data errors, which are inconsistent with Kerala's monsoon averages.

Warmest Zones: Due to heat retention and sun intensity, low elevations (CLASS 1-2) continuously recorded the highest maxima and minima.

Coldest Zones: Elevation-driven cooling trends were followed by minima and maxima, with high elevations (CLASS 5–6) remaining the coolest throughout the year.

Monsoon Disruptions: During the monsoon months, unusual cooling in mid- and low-elevation zones (CLASS 2, 4, 5) raised questions about normal climatic expectations and may have been the result of infrequent microclimatic phenomena or inconsistent data.

4.2.4. 2019

There were noticeable temperature variations in Kerala's elevation classes, which ranged from CLASS 1 (>0 meters) to CLASS 6 (2000–3000 meters). Lowland and coastal zones, represented by CLASS 2 (0–100 meters), maintained the highest monthly average temperatures, reaching a record high of 42.95°C in June, which was exceptionally high for the pre-monsoon season. In June, CLASS 3 (100–500 meters) recorded a temperature of 41.81°C, indicating strong solar heating in slightly higher inland regions. Tab.17 showcase the highest monthly average maxima on 2019.

2019 Wise Monthly LST	Altitude Average	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY		30.0215	34.61105	34.3633	30.45	29.0194	27.5522
FEBRUARY		33.4029	37.4137	37.3652	34.3244	31.6266	27.5121
MARCH		37.82142	39.909	40.25818	38.7966	38.85947	30.98619
APRIL		39.7366	41.2227	41.35	40.5711	37.0422	33.4236
MAY		36.37	40.4	40.13	38.8311	35.44	28.9542
JUNE		29.97	42.95	41.81	38.2775	33.8525	27.19
JULY		29.51	39.79	39.79	34.05	28.57	23.29
AUGUEST		30.43	34.83	33.67	37.55	32.21	23.31
SEPTEMBER		32.81	39.15	34.79	35.28	30.34	24.21
OCTOBER		31.5366	38.65	36.25	33.49	32.57	28.96
NOVEMBER		31.49	36.69	32.566	29.74	27.9585	24.876
DECEMBER		32.0427	35.285	32.8333	30.2269	27.55363	25.13545

Tab.17. highest monthly average LST on 2019 in Degree Celsius.

However, CLASS 6 (2000–3000 meters), which includes high-altitude areas, had the lowest average monthly temperature throughout the year, with July being the lowest at 14.97°C. These extremes demonstrate the strong inverse relationship between temperature and height while maintaining the environmental lapse rate. Tab.18 shows the lowest monthly average LST on 2019.

2019 Wise Average LST	Altitude Monthly	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY		26.0433	25.93	24.4973	21.213	18.7909	18.75727
FEBRUARY		28.1681	27.7338	26.4361	23.8825	20.7611	27.39316
MARCH		29.8211	27.816	27.75086	26.222	23.12	22.7573
APRIL		31.2122	29.69	27.425	24.74714	21.6677	21.81
MAY		25.23	23.87	25.18818	21.92	21.38	21.64789
JUNE		20.87	1.97	15.61	0.29	10.11	15.282
JULY		20.37	1.47	11.09	1.69	1.69	14.97
AUGUST		24.15	10.39	10.51	15.33	11.09	11.09
SEPTEMBER		24.97	13.99	19.61	13.01	7.71	16.07
OCTOBER		27.85	15.49	14.07	15.69	6.69	10.97
NOVEMBER		28.32	25.45	23.26714	21.138	16.94	17.3871
DECEMBER		27.7212	25.80286	23.0722	19.222	17.705	17.43

Tab.18. lowest monthly average LST on 2019 in Degree Celsius.

Seasonal shifts further emphasized altitude's role in Kerala's thermal dynamics:

Summer (March to May): Heat extremes were concentrated at lower elevations (CLASS 1–3), with CLASS 2 reaching a peak of 41.22°C in April. This is a typical pattern as solar radiation increases before the monsoon. In April, the average temperature at higher elevations (CLASS 6) remained considerably lower at 21.81°C, indicating the moderating influence of altitude.

Monsoon (June to September): Although CLASS 2 recorded an incredible 42.95°C in June, most likely as a result of pre-monsoon heat that persisted, the following months displayed unpredictable trends. For example, in July, CLASS 2 recorded an extremely low temperature of 1.47°C, which is not consistent with Kerala's tropical monsoon environment. This could be the result of malfunctioning sensors. CLASS 6 stayed chilly, falling to 11.09°C in August, which is consistent with high-altitude climate standards.

Winter (December–February): The coldest temperatures were found at higher elevations (CLASS 5–6), with CLASS 5 dropping to 16.94°C and CLASS 6 averaging 17.43°C. Winters were milder at lower elevations (CLASS 1–2), highlighting the stabilizing effect of lowland warmth.

CLASS 2 (0–100 meters) emerged as the hottest zone due to intense solar heating in lowland regions, while CLASS 6 (2000–3000 meters) remained the coldest, reflecting the cooling effects of higher altitudes. These elevation-driven temperature patterns in Kerala during 9 demonstrated a clear inverse relationship between altitude and thermal conditions. The topographical diversity of Kerala is highlighted by this gradient, where elevation significantly influences regional climates. The data did, however, show several noteworthy anomalies, such as the unrealistic 1.47°C in July for CLASS 2, which was a sharp departure from Kerala's monsoon standards, which usually include warm, humid weather. Such anomalies probably point to data entry mistakes or sensor problems, which calls for more investigation. Across lower elevations (CLASS 1–3), summer heat peaked seasonally in April–May, coinciding with pre-monsoon solar intensity. June through July, the monsoon months, were volatile, especially in CLASS 2, where contradictory extremes, such the high of 42.95°C in June and the unusual low of 1.47°C in July, point to inconsistent data or isolated microclimatic disturbances. This supports the moderating effect of altitude on seasonal variability. In contrast, winter temperatures stayed most constant at higher elevations, with CLASS 6 exhibiting the fewest variations (e.g., 17.43°C in December). The aforementioned patterns underscore the necessity of verifying data from the monsoon season while reinforcing the fundamental function of elevation in Kerala's thermal dynamics.

4.2.5. 2024

The elevation gradient of Kerala, which ranges from CLASS 1 (>0 meters) to CLASS 6 (2000–3000 meters), showed a definite negative correlation between temperature and altitude. The warmest zones were CLASS 2 (0–100 meters) and CLASS 3 (100–500 meters), with both reaching a record-breaking 46.13°C in May for Kerala's tropical environment. In lowland areas, this intense heat corresponds with pre-monsoon sun intensity. However, in line with high-altitude cooling. Tab.19 shows the higher monthly average LST on 2024.

2024 Altitude Wise Monthly Average Maximum LST	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY	29.7313	34.4485	31.70176	29.0522	27.08	23.4382
FEBRUARY	33.411	35.5928	34.6136	31.68047	29.5481	26.8816
MARCH	35.1627	36.35	37.056	35.5658	34.1583	29.9941
APRIL	35.6285	40.7575	39.7588	39.0793	34.5537	30.5921
MAY	38.05	46.13	46.13	38.5233	36.41	30.79
JUNE	29.94	38.41	34.43	34.43	31.29	31.85
JULY	29.19	33.47	29.93	36.55	32.65	25.69
AUGUST	30.89	36.71	34.81	35.81	31.61	24.85
SEPTEMBER	30.33	35.27	32.01	34.35	31.142	23.1666
OCTOBER	32.59	35.45	31.88	29.09	27.11	23.83
NOVEMBER	30.16428	34.71	31.31	27.3488	26.018	21.8614
DECEMBER	29.966	32.94	30.1209	27.7585	25.975	23.12294

Tab.19. highest monthly average LST on 2024 in Degree Celsius

CLASS 6 (2000–3000 meters) had the lowest maximum temperatures, averaging 23.12°C in December. A similar pattern was seen in minimum temperatures. Lows of 7.75°C in June and 11.01°C in July, which indicate the ongoing coolness of elevated zones, made CLASS 6 the coldest month of the year. This trend was broken by anomalies, though: in August, CLASS 2 recorded an unrealistic -1.61°C, which is out of line with Kerala's monsoon environment and most likely the result of a sensor or data entry error. Lowlands retained more heat, as seen by milder low temperatures (e.g., 26.67°C in January) at lower elevations, especially CLASS 1 (>0 meters). Additional outliers that needed to be verified for correctness were CLASS 4 (500–1000 meters) at 3.75°C in July and CLASS 5 (1000–2000 meters) at 4.63°C in August. tab.20 shows the lowest monthly minima of 2024.

2024 Altitude Wise Monthly Average Minimum LST	CLASS 1	CLASS 2	CLASS 3	CLASS 4	CLASS 5	CLASS 6
JANUARY	26.67	26.02	24.195	21.5285	18.1566	17.4025
FEBRUARY	27.7091	27.488	25.4671	23.5919	19.6386	19.2066
MARCH	29.468	28.336	27.2512	25.3891	22.7012	22.2152
APRIL	30.3816	27.83	26.8447	24.5313	22.615	22.4131
MAY	21.57	13.01	12.89	8.57	10.37	17.16
JUNE	23.02	5.39	18.41	7.85	7.75	7.75
JULY	22.11	7.91	7.37	3.75	6.11	11.01
AUGUST	23.0366	-1.6099	6.21	15.19	4.63	8.57
SEPTEMBER	24.14	18.36	20.63	18.95	11.81	13.0328
OCTOBER	26.85	18.41	19.97	19.9	15.43	15.92
NOVEMBER	27.194	23.4100	21.74	16.5933	13.37	14.80333
DECEMBER	27.0825	25.37	23.2683	21.1347	17.99941	17.7876

Tab.20. lowest monthly average LST on 2024 in Degree Celsius.

4.3. Seasonal Variations

4.3.1. Summer (March–May): May was the hottest month, with pre-monsoon aridity causing CLASS 2 and 3 temperatures to reach 46.13°C. Although the minimum temperatures in CLASS 5–6 still averaged 17–22°C, lower elevations (CLASS 1–3) continuously outperformed higher zones (CLASS 4–6) in both maxima and minima. May minima that abruptly drop (e.g., 8.57°C for CLASS 4) may indicate anomalies in the data or localized cooling.

4.3.2. Monsoon (June–September): Unpredictable readings plagued the season, especially in CLASS 2, which reported unlikely lows of 5.39°C in June and -1.61°C in August—values that were outside of Kerala's typical humid

monsoon conditions. While mid-altitudes (CLASS 2–4) had fluctuating maxima, such as 38.41°C in June and 36.55°C in July, which were probably impacted by cloud cover dynamics, higher elevations (CLASS 6) maintained stable coolness (e.g., 8.57°C in August).

4.3.3. Winter (December to February): CLASS 1–2 stayed over 25°C (e.g., 26.02°C in January), but CLASS 6 maintained its position as the coldest zone with minimum averaging 17.4–23.1°C. Altitude's stabilizing function throughout the colder months was further supported by the minor changes in winter temperatures.

Lower elevations (CLASS 1–3: 0–500 meters) consistently recorded higher temperatures due to solar heating and heat retention, while higher elevations (CLASS 4–6: 500–3000 meters) remained cooler, adhering to the environmental lapse rate, according to the 2024 temperature analysis, which confirmed Kerala's elevation-driven thermal hierarchy. But the dataset also showed surprising abnormalities, as the improbable -1.61°C in CLASS 2 (0–100 meters) in August and sudden May minima reductions (e.g., 8.57°C in CLASS 4), which were probably caused by sensor failures. Seasonal variations also highlighted climate extremes: monsoon months showed unpredictable volatility in CLASS 2, with values like 5.39°C in June defying Kerala's humid norms and underscoring the need for thorough data validation, while summer heatwaves in May caused CLASS 2–3 to soar to 46.13°C, highlighting lowland vulnerability to pre-monsoon aridity. Higher elevations (CLASS 6) had winter stability, with temperatures ranging from 17.4 to 23.1°C. This was in sharp contrast to lowland zones (CLASS 1–2), which experienced mild lows that hardly ever dropped below 25°C.

CHAPTER 5

5.SUMMARY AND CONCLUSION

Thus, the LST trend in Kerala during the past 20 years is clearly shown by this analysis. Looking at latitude, it is evident that 10–12° latitude displaying the highest peaks in each year, indicating that this region—which includes districts like Palakkad, Thrissur, Ernakulam, etc.—dominates Kerala's highest LST. The main causes of this greater temperature can be urbanization, decreased vegetation density (it's about lower NDVI in some years), and intense solar radiation during the pre-monsoon summers. It is likely that latitudes 12°–13° are the coolest. This latitude class, which includes districts like Kannur and Kasargod, has stable low temperatures and moderate temperatures. Mostly due to the high NDVI values (the 2004 NDVI was 0.9857) for the majority of the year. Additionally, there were occasional unpredictable monsoon lows in this region (for example, -4.83°C in July 2009, perhaps due to sensor mistake) or cloudy condition.

Based on the trend in height, class 1-3 (0-500 altitude) consistently recorded the greatest maximum temperature (e.g., 46.13°C in May 2024) and the highest peak temperature. This class will mostly include coastal cities and lowland areas, such as Kochi, Calicut, Trivandrum, Alappuzha, and so on. For the majority of the year, the class 5-6 (1000–3000m) reported steady minima (e.g., 7.75°C in June 2024). This category primarily includes hilly areas like Munnar, Nelliampathi, and Attapadi etc.

5.1. Temperature Variation In 20 years;

This is evident in the lowland warning Trend, where the peak summer temperature rose by around 3°C (for example, 43.17°C in 2004 and -46.13°C in 2024) and monsoon volatility increased. The irregular low temperatures (for example, -1.61°C in August 2024) are probably caused by microclimates or sensor mistake. At mid-altitudes (500-1000 meters), the monsoon maximum temperatures have risen (from 37.07°C in 2004 to 38.57°C in 2014, for example). Heat mapping beneath clouds is suggested. Attitudes between 2000 and 3000 meters are where the cooling trend is most noticeable. The winter minimum temperature is constant at higher elevations (17:43°C in December 2019). However, summer temperatures barely increased (for example, to 21.81°C in April 2019).

Region Type	Average Max Temp (2004–2024)	Average Min Temp (2004–2024)	Temp Variation Magnitude
Lowlands (0–500m)	42.5°C (Summer)	25.2°C (Winter)	+3.5°C (Summer peaks)
Mid-Altitudes (500–1000m)	38.1°C (Monsoon)	18.7°C (Winter)	+1.8°C (Monsoon)
High Altitudes (2000–3000m)	24.3°C (Winter)	9.8°C (Monsoon)	-0.5°C (Stable)

Fig.9. average temperature by region

5.2. Key finding:

According to the data, the lowlands are experiencing an increase in temperature. (class 1-3), which includes urban and coastal areas, experienced highest warming due to decreased vegetation and urbanization. There has been monsoon volatility. Erratic lows have been seen at lower elevations (e.g.: -1.61°C in 2024). Data inaccuracies are suggested, but these might be a reflection of localized cooling in valleys. Class 6 reported the High-Altitude Stability, which is essential for biodiversity in the face of long-term warming, and maintained steady lows.

Why 10-11° latitude experience higher LST maxima compared to other latitudes? ;

From this study, it's clearly depicting that 10-12° latitude dominating in maximum average temperature, and also dominating in the peak maxima also. Apparently 8-9° latitude is near to the equator, but 10-11° latitude has the highest LST maxima, which proves that area of Kerala, includes district of Palakkad, Thrissur, and some portion of Ernakulam, Malappuram, and Idukki. There are some reasons for the higher LST's in that area. The main reason for that is the **Palakkad Gap**, which is a natural break in the Western Ghats mountain range. This gap allows the movement of air between the coastal plains and the higher elevation of the Western Ghats. This gap facilitates the entry of warm, dry wind from Tamil Nadu into Palakkad, which contribute to the higher temperature. The gap also allows moisture from the southwest monsoon to escape the region, leading to lower humidity level compared to other parts of Kerala, where monsoon winds are more restricted.

5.3. FURTHER STUDIES NEEDED

The study draws attention to the discovery of anomalous monsoon temperature patterns in Kerala, specifically highlighting the existence of negative land surface temperature (LST) readings that depart from the conventional climatic trends of the area. These irregularities might be explained by variations in the microclimate or errors in the sensors. Given Kerala's coastal terrain and the dynamic role of humidity impacted by various climatic factors, it is advised to use humidity mapping by remote sensing in order to enrich the research. This method may help identify changes in the local microclimate. Furthermore, incorporating trends in urbanization, land-use/land-cover (LULC) dynamics, and rainfall patterns will improve the accuracy of LST models and advance our knowledge of microclimatic fluctuations. To detect and reduce possible sensor differences, data validation procedures such as error filtering and quality checks are essential.

Field surveys combined with remote sensing would provide ground-truth insights into the conditions of the study region and provide solid, actionable conclusions for future research, further validating the reliability of the data. All of these tactics, which include thorough data validation, multi-parameter mapping, and on-site verification, work together to offer a thorough framework for deciphering Kerala's complicated climate and resolving uncertainty in observed anomalies.

CHAPTER 6

6. Bibliography

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